Lorentzian Einstein–Cartan Reduction on Compact Homogeneous Geometries: Constraints and Inhomogeneous Perturbations

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Version 2
Revised: 2026-11-08
First release: 2026-06-14

Abstract

We study inhomogeneous constraints in source-free Lorentzian Einstein–Cartan theory with constant complete Nieh–Yan coupling on fixed compact geometries. The regular homogeneous metric counts remain 5, 3, 2 and 2 pairs on the torus, Nil, Sol and the three-sphere, with symmetric-background exceptions treated separately. For nonzero flat-torus modes, connection-retaining and metric reductions agree in the action and reduced two-form. Under the stated compact CMC hypotheses, the Arms–Marsden–Moncrief theorem makes the four Taub conditions necessary and sufficient for local integrability; explicit standing-wave and nonparallel two-wave corrections satisfy the local vacuum equations through second order. On selected time-dependent curved vacuum backgrounds, we construct closed tensor blocks on LRS Nil, infinite dual-orbit systems on Sol, and low Wigner blocks on Taub, with gauges, canonical reductions and reconstruction. On the fixed Sol background, the no-KID theorem gives local integrability of all real full linearized constraint data in the stated Sobolev class, including admissible Gowdy and common-core dual-orbit data. Quadratic KID conditions on Nil and Taub remain necessary conditions only. Finite-time examples include an independent Lorenz evolution on one Sol orbit, with interior spatial, window and time convergence comparisons. Convergence in the reduced fourth-order form, whole-line numerical error bounds, general stability, nonlinear closure within a fixed sector and lifting sufficiency on Nil and Taub are not established.

1 Question and results

We study the passage from homogeneous constraints to inhomogeneous perturbations on fixed compact spaces in source-free Lorentzian Einstein–Cartan (EC) theory with constant complete Nieh–Yan (NY) coupling. Elimination of the independent connection, the linear gauge quotient, and tangency to an exact vacuum family are distinct stages of the same variational problem.

We use the action, quotient and boundary conventions of Papers01–05 [1, 2, 3, 4, 5, 6]. Paper06 treated the full equations of common-scale families, the homogeneous volume–shape cone and its exact homogeneous realization [7]. Here these results provide the starting point for spatially dependent fields, closed harmonic blocks and background-dependent integral conditions.

First, the regular homogeneous metric strata have 5/3/2/2 canonical pairs on T^3/Nil/Sol/S^3. The static torus and symmetric Taub points are treated separately. For nonzero flat-torus modes, direct reduction before and after Lorentz gauge fixing and connection elimination gives the same quadratic dynamics and reduced two-form.

Second, the four flat-torus Taub conditions are necessary and sufficient for local integrability under the hypotheses of the Arms–Marsden–Moncrief theorem [18]. We give explicit second-order corrections for a standing wave with homogeneous expansion and retain all thirteen generated

wavevectors for two nonparallel waves. The theorem application and the finite-order construction have different logical roles.

Third, we reduce inhomogeneous constraints on selected time-dependent vacuum backgrounds of Nil, Sol and S^3. We treat the central-zero and central-nonzero Nil sectors, fiber-independent Sol modes and general nonzero dual orbits, and four low Wigner labels on S^3, with complete reconstruction for $j = 3/2$. Finite-time examples include an independent Lorenz evolution on Sol with interior convergence comparisons.

Two propagating field species, the total canonical pairs in a representation block, and the homogeneous regular-stratum count are different quantities. For Nil and Taub we establish quadratic necessary KID conditions. On the fixed Sol background, absence of KIDs and the existing theorem give local integrability of all full linearized constraint data in the stated Sobolev class (Section 7 and Appendix G). All-time stability and a complete physical spectrum are not established. Closed forms, finite metric paths and spatial Laplacians remain separate geometric data in Appendix E.

Auxiliary connection elimination, the two gravitational polarizations and linearization stability belong to established theory [8, 13, 17, 18, 19]. The concrete content here is the explicit constraints, reductions, reconstructions and second-order corrections for the stated action and quotients, with their counting units and domains.

2 Action, geometry and boundary conditions

The signature is $(-,+,+,+)$, the orientation is positive e^{0123} and $\epsilon_{0123} = +1$. The coframe is nondegenerate, spatial slices are spacelike and the independent connection is Lorentz-valued. The NY coupling θ is constant and $\kappa^2 > 0$. Matter, spin sources, a cosmological constant and curvature-square terms are absent.

$$\begin{aligned} T^a &= de^a + \omega^a{}_b \wedge e^b, \qquad R^a{}_b = d\omega^a{}_b + \omega^a{}_c \wedge \omega^c{}_b, \\ Q &= e^a \wedge T_a, \qquad dQ = T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab}, \\ S &= \frac{1}{4\kappa^2} \int \epsilon_{abcd} e^a \wedge e^b \wedge R^{cd} + \theta \int dQ + S_\partial. \end{aligned} \tag{A-01}$$

With all antisymmetric indices summed, the epsilon four-form equals $2R_{\mathrm{EC}} e^{0123}$. The first term is therefore $R_{\mathrm{EC}}/(2\kappa^2)$. This normalization is used consistently in canonical momenta and kinetic coefficients.

$$d\sigma^i = \frac{1}{2} n_{ij} \epsilon_{jkl} \sigma^k \wedge \sigma^l, \quad n^T = n, \quad \mathcal{V}_\Sigma = \int_\Sigma \sigma^{123}, \quad g = g_{ij}(t) \sigma^i \sigma^j. \tag{G-01}$$

The class-A structure matrix n is fixed, while g is any positive-definite symmetric matrix. Constant $SO(3)$ rotations preserve the rank of n. The fixed quotients and normalizations are listed below; q is the physical spherical radius. Appendix A specifies the identifications.

Geometry	Eigenvalues of n	$\mathcal{V}_\Sigma$	${}^{(3)}Rq^2$
T^3	$(0,0,0)$	$(2\pi)^3$	0
Nil	$(0,0,-1)$	$(2\pi)^4$	$-1/2$
Sol	$(0,1,-1)$	$\log((3+\sqrt{5})/2)$	-2
S^3	$(2,2,2)$	$2\pi^2$	6

2.1 The complete connection equation

$$\epsilon_{abcd}e^c \wedge T^d = 0 \quad \Longrightarrow \quad T^a = 0, \qquad \omega = \omega_{\rm LC}[e]. \tag{A-02}$$

Constant complete NY does not contribute to this bulk equation. Inverting epsilon gives $e^a \wedge T^b - e^b \wedge T^a = 0$. With $t = \sum_a \iota_a T^a$, contraction yields $T^b = -e^b \wedge t$, then $t = -3t$, hence $t = T = 0$. This proof uses no AX/VT/MX restriction. Appendix B supplies a 24-component algebraic control.

Writing $K = \omega - \omega_{\rm LC}$ splits the EC scalar into its LC value, an algebraic quadratic form in K, and a divergence. After transferring the divergence to the endpoints, K is an invertible auxiliary field. Because $\omega_{\rm LC}$ depends on coframe derivatives, canonical equivalence is not assumed from a point transformation; the Euler chain rule and endpoint one-form establish the correspondence.

2.2 Boundary representatives

At the two spacelike time boundaries we include the LC GHY term with outward-normal signs and $\epsilon_n = -1$. The bulk representative S_0 additionally subtracts the contortion-divergence and NY currents at the endpoints and is varied with fixed induced metric. A representative retaining these currents is compared with corresponding data fixing their endpoint variation. No boundary sources are introduced.

$$\begin{aligned} S_{\rm GHY} &= \kappa^{-2}\sum_{i,f}\epsilon_n \int_\Sigma \sqrt{h}\, K_\partial, \qquad \epsilon_n = -1, \\ S_{\rm keep} &= S_0 + [B_{\rm end}]_i^f, \quad \Theta_{\rm keep} = \Theta_0 + \delta B_{\rm end}, \quad \Omega_{\rm keep} = \Omega_0. \end{aligned} \tag{A-03}$$

In the common-scale two-amplitude family, geometric torsion is $T^i = (V/3)e^{0i} + (2\eta/q)e^{jk}$ for cyclic triples. With $D = N^{-1}d/dt$ and $H = Dq/q$, the divergence and endpoint function are as follows. The HV term cannot be discarded alone.

$$\begin{aligned} R_{\rm EC} - R_{\rm LC} &= -6\eta^2/q^2 + 2V^2/3 - 2DV - 6HV, \\ B_{\rm end} &= -\mathcal{V}_\Sigma q^3 V/\kappa^2 + 6\theta\mathcal{V}_\Sigma q^2\eta, \\ P_i &= p_i + \partial_i B_{\rm end}. \end{aligned} \tag{A-04}$$

Fixing q, η, V at the endpoints sets $\delta B_{\rm end} = 0$ in this representative; lapse and scale velocity need not be fixed. On a smooth complete-connection shell, $\eta = V = 0$ also holds at the endpoints. Removing a divergence does not create a kinetic term, and a time-linear current is not a propagating mode.

3 Backgrounds and the six-component homogeneous action

$$\begin{aligned} c = \frac{\mathcal{V}_\Sigma}{2\kappa^2}, \quad d = \det g, \quad v = \dot g - \mathcal{L}_\beta g, \quad K = \frac{1}{2N}g^{-1}v, \\ P(n,g) = \frac{1}{2}[\mathrm{tr}(ng)]^2 - \mathrm{tr}(ngng), \qquad {}^{(3)}R = \frac{P(n,g)}{d}, \\ L_0 = c\sqrt{d}\left[\frac{\mathrm{tr}((g^{-1}v)^2) - [\mathrm{tr}(g^{-1}v)]^2}{4N} + N\frac{P(n,g)}{d}\right]. \end{aligned} \tag{H-01}$$

This action follows after solving the independent connection equation and treating the boundary. All six metric components, lapse and three shifts are varied before gauge choices. The shifts use

invariant vector fields descending to the fixed quotient. Unimodularity removes the invariant-volume divergence, and this complete homogeneous variation retains the homogeneous Einstein equations [15]. Counts obtained by imposing a diagonal metric beforehand are not the counts studied here.

The curvature formula follows from $g = E^{\mathsf{T}}E$, the orthonormal structure matrix $m = EnE^{\mathsf{T}}/\det E$, and $R = \frac{1}{2}(\mathrm{tr}m)^2 - \mathrm{tr}(m^2)$. The kinetic term is $K_{ij}K^{ij} - K^2$. The metric Legendre map is invertible for positive-definite g; the six-velocity DeWitt form has five positive and one negative directions. This signature alone is not a physical ghost test.

$$\begin{aligned} \pi^{ij} &= c\sqrt{d}\,(K^{ij} - (\mathrm{tr}K)g^{ij}), \qquad T(g,\pi) = \mathrm{tr}(\pi g \pi g) - \tfrac{1}{2}[\mathrm{tr}(\pi g)]^2, \\ \mathcal{H} &= \frac{T - c^2 P}{c\sqrt{d}}, \qquad F = c\sqrt{d}\,\mathcal{H} = T - c^2 P. \end{aligned} \tag{H-02}$$

The symmetric π is spatially integrated, with all indices summed in $\pi^{ij}\delta g_{ij}$. The momentum conjugate to an independent off-diagonal coordinate is $2\pi^{ij}$. Replacing $\mathcal{H}$ by F multiplies the constraint by the strictly positive $c\sqrt{d}$ and rescales its Hamiltonian direction on the zero set by the same nonzero factor.

3.1 Testing common-scale backgrounds

The common-scale background test of Paper06 is recalled in (B-01) as the starting point for the constraint analysis here [7].

$$\begin{aligned} g = q^2 I, \quad 6\chi = \tfrac{1}{2}(\mathrm{tr}n)^2 - \mathrm{tr}(n^2), \quad X = 2n^2 - (\mathrm{tr}n)n + 4\chi I, \\ G_{00} = 3(H^2 + \chi/q^2), \qquad G_{0i} = 0, \\ \tfrac{1}{3}\sum_i G_{ii} = -(2DH + 3H^2 + \chi/q^2), \qquad (G_{ij})_{\mathrm{TF}} = X_{ij}/q^2. \end{aligned} \tag{B-01}$$

The complete connection shell gives the vacuum Einstein equations. A real common-scale reduced trajectory satisfies $(Dq)^2 + \chi = 0$, but X remains for Nil and Sol. Standard Nil has $X = \mathrm{diag}(-1,-1,2)/3$ and Sol has $X = \mathrm{diag}(-4,2,2)/3$. The static torus is a full solution; the round sphere admits no real common-scale vacuum trajectory in this model.

The quadratic action about a static sphere is consequently an off-shell algebraic control, not a vacuum particle spectrum. The same caution applies to shape-frozen Nil and Sol trajectories. General anisotropic vacuum solutions are not excluded. The regular strata in Section 5 use general g and momenta; Appendix C gives constraint-satisfying representatives.

Perturbations $e = \bar{e} + \varepsilon f$ and $\omega = \bar{\omega} + \varepsilon a$ must be expanded with the same grading. A nonzero first variation is not discarded to declare an on-shell background; background residuals in Noether identities remain relevant. Equal coframe scales on Nil or Sol do not provide spatial isotropy or guarantee separate scalar, vector and tensor closure.

3.2 Time-dependent vacuum backgrounds for perturbations

For curved geometries we use the following time-dependent vacuum backgrounds instead of a common-scale reference metric. G is the spatial metric in the fixed invariant coframe of Appendix A. Appendix F gives their normalization, quotient identifications and verification of the full Ricci tensor and ADM constraints.

$$
\begin{aligned}
\text{Nil}: &\quad G=\operatorname{diag}(A,A,16u^2/(9A)),\quad N=3\sqrt{A},\quad A=1+u^4,\\
\text{Sol}: &\quad G=\operatorname{diag}(F,t,t),\quad N=\sqrt{F},\quad F=e^{t^2}/\sqrt{t},\\
S^3: &\quad G=\operatorname{diag}(4(1+\tau^2),4(1+\tau^2),16(1-\tau^2)/(1+\tau^2)),\\
&\quad N=\sqrt{(1+\tau^2)/(1-\tau^2)}.
\end{aligned}
\tag{BF-01}
$$

We use $1\le u\le 2$ on Nil, $1\le t\le 2$ on Sol, and $1/4\le\tau\le 1/2$ in the Taub region. Throughout these intervals $N>0$ and G is positive definite. The Taub parameters are $\ell=1$, $m=0$, away from both the Cauchy horizon and the time-symmetric slice. Lifting with the LC connection gives vacuum backgrounds of the same EC theory [26, 27, 28].

4 Connection elimination and constraint preservation

We compare the connection-retaining and metric descriptions at the same perturbative order with the same action and endpoint data. The derivative-dependent LC substitution is not asserted to be a point canonical transformation on the unreduced space [13].

$$
\frac{d\phi_A}{dt}=\{\phi_A,H_c\}+\{\phi_A,\phi_B\}\lambda^B\simeq 0,\qquad H_T=H_c+\lambda^B\phi_B. \tag{D-01}
$$

In primary preservation, the image of the Poisson matrix fixes multipliers; only its left kernel produces new constraints. The original primary multipliers are retained at every consistency stage. Appendix C gives the scalar actions (D-02)–(D-03) and their closure.

4.1 Direct comparison for a nonzero flat mode

Description	Configurations	First class	Second class	Pairs
Lorentz retained	40	20	36	2
Lorentz fixed	34	8	48	2
Metric	10	8	0	2

The table is for one real Fourier quadrature. Retaining all 24 connection components and symbolic $k\neq 0$, the reduced action, endpoint-derived two-form and the kinetic and gradient coefficients of both polarizations agree with (W-02) in all three descriptions. Including both real quadratures gives four real canonical pairs.

$$
\Theta_{\text{ret}}\big|_{\omega_{\text{LC}}}-\Theta_{\text{met}}=\delta B,\qquad \Omega_{\text{ret}}\big|_{\omega_{\text{LC}}}-\Omega_{\text{met}}=\delta^2 B=0. \tag{D-05}
$$

B includes the Lorentz total derivative and the complete NY endpoint. The current mixing the two real quadratures is retained, combining both NY terms from the same connection. For curved backgrounds we use the full Cartan solution, the auxiliary quadratic form and endpoint identities. We do not claim a new complete unreduced Dirac chain for every curved block; Appendix B specifies the directly verified scope.

5 Regular homogeneous constraints and the static torus

$$
\begin{aligned}
(A_k)^i{}_j&=n_{iz}\epsilon_{zkj},\qquad \mathcal{L}_\beta g=\beta^k(A_k^Tg+gA_k),\\
J_k&=\operatorname{tr}[\pi(A_k^Tg+gA_k)],\\
\{J_i,F\}&=0,\qquad \{J_i,J_j\}=-\epsilon_{ijk}n_kJ_k\quad (n=\operatorname{diag}(n_1,n_2,n_3)).
\end{aligned}
\tag{H-03}
$$

The symplectic form is $\Omega = \sum_{ij} \delta\pi^{ij} \wedge \delta g_{ij}$. After removing the lapse and shift primary multiplier pairs, F and the independent J_k generate first-class constraints. Internal Lorentz directions have already been quotiented in the metric description after solving torsion. Discrete quotient automorphisms are not subtracted as continuous gauge directions.

$$r = \operatorname{rank} dJ, \qquad \operatorname{rank} d(F, J) = r + 1, \qquad N_{\mathrm{hom}} = 6 - r - 1. \tag{H-04}$$

Equation (H-04) is a local count on the regular open stratum of the constraint surface. Appendix C gives constraint-satisfying representatives and nonzero minors. The span of the A_k bounds the number of independent J_k, and the representative minors attain that bound. Degenerate strata require a separate regularity check.

Geometry	$\operatorname{rank} dJ$	$\operatorname{rank} d(F, J)$
T^3	0	1
Nil	2	3
Sol	3	4
S^3	3	4

The counts 5/3/2/2 concern the finite-dimensional system of all symmetric homogeneous metrics on the specified fixed quotients. They are not graviton polarizations per spacetime point, Teichmüller dimensions for arbitrary lattices, or zero-frequency counts at each background. Their objects and values differ from the 3/2/1/0 closed-one-form dimensions.

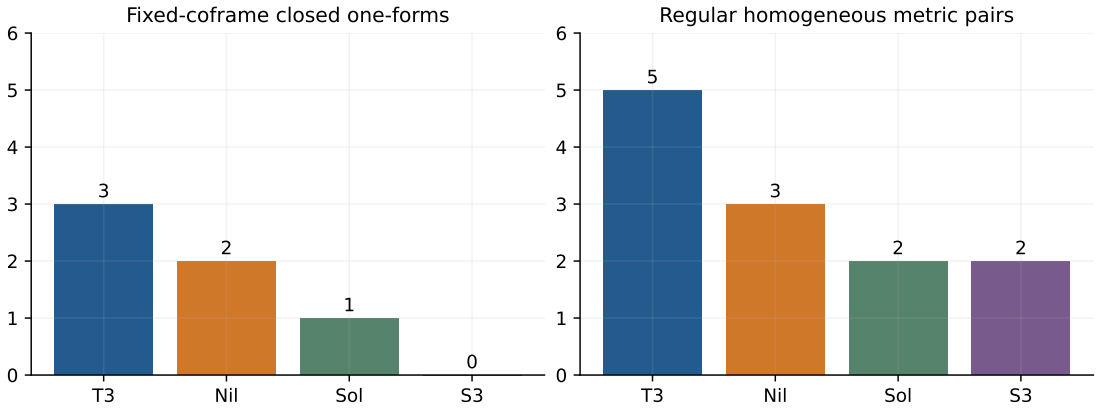


Figure 1: Constant closed-one-form dimensions in the specified coframes and homogeneous pair counts on regular constraint strata. The static torus is not on the latter regular stratum.

5.1 The singular static torus

The homogeneous mixed cone and exact family of Paper06, Section 5 and Appendix D, are placed here on the singular stratum of the Hamiltonian constraint F [7]. Sections 6–7 add nonzero Fourier components to this setting.

At $n = 0$, $\pi = 0$, F is quadratic in momenta and $dF = 0$. The quadratic action alone retains six metric pairs because its lapse equation vanishes. The second-order coefficient of the original nonlinear Hamiltonian constraint instead imposes the following condition, where h is the first-order mean expansion and Σ is tracefree.

$$K = \varepsilon(hI + \Sigma) + O(\varepsilon^2), \qquad \mathrm{tr}\Sigma = 0, \qquad \mathrm{tr}(\Sigma^2) = 6h^2. \tag{T-01}$$

This cone is not a vector space. Its pure-scale slice requires $h = 0$, while nontrivial simultaneous shape and volume directions exist. Replacing $p_q^2 = 0$ by $p_q = 0$ changes its differential and gauge generator and is not an equivalent regularization.

$$\begin{aligned} g_{ii}(\tau,\varepsilon) &= q_0^2(1+3\varepsilon h\tau)^{2p_i}, \quad p_i = \frac{1}{3} + \frac{\sigma_i}{3h}, \\ \sum_i \sigma_i = 0, &\quad \sum_i \sigma_i^2 = 6h^2, \quad \sum_i p_i = \sum_i p_i^2 = 1. \end{aligned} \tag{T-02}$$

For $h \neq 0$, (T-02) is an exact vacuum solution on $1 + 3\varepsilon h\tau > 0$, and its derivative in ε realizes the cone direction. A general symmetric Σ is recovered by a constant orthogonal matrix representation. If $h = 0$, then $\Sigma = 0$ and constant metric deformations remain. This explicit homogeneous construction is distinct from the theorem application for spatially dependent waves in Section 7.

The static flat-torus metric family is locally six-dimensional SPD(3), or five-dimensional at fixed volume, with residual discrete lattice identifications. Its pulled-back symplectic form vanishes because $\pi = 0$; this does not make the whole family gauge.

The closed-one-form dimensions of Paper04 and frame comparisons of Paper06 serve as geometric data specifying the homogeneous truncation [5, 7]. Obtaining a reduced metric kernel from those dimensions would additionally require a map to perturbation fields compatible with constraints and gauge quotient. Appendix E collects the preceding geometric results in the present normalization.

5.2 Linear rank exceptions at Taub points

At the S^3 Taub background (BF-01), the homogeneous momentum rank is two and the full constraint rank is three. Axial $U(1)$ symmetry lowers these from the generic ranks three and four, giving a three-pair linear gauge quotient. This does not replace the two-pair regular-stratum result (H-04). Quadratic KID conditions remain, so the linear quotient is not identified with a smooth nonlinear physical phase space.

The ℓ, m tangent directions of the Taub family span one nondegenerate linear pair, while background time displacement is pure gauge. Further KID exceptions occur at low inhomogeneous labels and are counted with their representation multiplicities in Section 6.

6 Constraint reduction of inhomogeneous perturbations

6.1 Constraints, time dependence and real-field conventions

In each block the six metric components, lapse and three shifts are retained before variation. Conjugate sectors are restored for real fields; independent off-diagonal momenta are $p_{ij} = 2\pi^{ij}$. With $z = (h, p)$, $\Omega = -J$, gauge matrix G and embedding E, the common reduction is

$$\begin{aligned} &CJC^\dagger = 0, \qquad (\dot{C} + CJH)\ker C = 0, \qquad CE = GE = 0, \\ &E^\dagger \Omega E = \Omega_r, \qquad H_r = E^\dagger H E - E^\dagger \Omega \dot{E}, \\ &\dot{y} = J_r H_r y, \qquad z = Ey. \end{aligned} \tag{R-01}$$

The adjoint includes spatial integration by parts when needed. Derivatives of time-dependent constraints and embeddings are retained. Lapse and shift are reconstructed together with the reduced fields. Appendices F–G give bases, closure, gauges, nonzero minors and reconstruction.

6.2 The fixed flat torus

The background $N_0 = 1$, $g_0 = q_0^2 I$ is an exact vacuum. For counting set $c = q_0 = 1$, express the nonzero wavevector along z, and use unit-mean-square real harmonics: cosine for metric and lapse and sine for shift.

$$\begin{aligned} v_{ij} &= \dot{h}_{ij} - k(\beta_i \delta_{j3} + \beta_j \delta_{i3}), \\ L_k^{[2]} &= \tfrac{1}{4}[\mathrm{tr}(v^2) - (\mathrm{tr}v)^2] + \mathcal{R}^{[2]}(h) + \alpha k^2 (h_{11} + h_{22}), \\ \mathcal{R}^{[2]} &= \delta^{ij}(\Gamma^{k(1)}{}_{i\ell}\Gamma^{\ell(1)}{}_{jk} - \Gamma^{k(1)}{}_{ij}\Gamma^{\ell(1)}{}_{k\ell}). \end{aligned} \tag{W-01}$$

The four primary and four secondary constraints are first class. The secondaries are proportional to $p_{13} = p_{23} = p_{33} = 0$ and $h_{11} + h_{22} = 0$, leaving two pairs in the gauge quotient. The polarization-frame rotation need not be a lattice automorphism. No division by nonzero k is continued to $k = 0$; the original quadratic action there retains six metric pairs.

$$\begin{aligned} & h_{11} = A, \quad h_{22} = -A, \quad h_{12} = B, \quad h_{i3} = 0, \\ & L_{\mathrm{TT}}^{[2]} = \tfrac{1}{2}(\dot{A}^2 + \dot{B}^2 - k^2(A^2 + B^2)), \\ & S_{\mathrm{TT}}^{[2]} = \frac{\mathcal{V}_\Sigma q_0^3}{8\kappa^2} \int d\tau \left\langle \dot{\gamma}_{ij}\dot{\gamma}_{ij} - q_0^{-2}\partial_a\gamma_{ij}\partial_a\gamma_{ij} \right\rangle. \end{aligned} \tag{W-02}$$

γ is a zero-mean TT field. The two polarizations in one real quadrature have positive kinetic terms and $\omega^2 = |k|^2/q_0^2$ on the static torus. A complete real $\pm k$ block has four pairs. This count is distinct from the integrated conditions for exact families in Section 7.

6.3 Nil: zero and nonzero central sectors

Central index $m = 0$ is inhomogeneous when $(n_1, n_2) \neq 0$. On the LRS background the covector is aligned to $(k, 0)$ in the tensor frame, with $k^2 = n_1^2 + n_2^2$; this does not rotate or reidentify the lattice. Put $R = 4u/3$ and $\zeta = 16u^2/[9(1 + u^4)^2]$.

$$L^{[2]} = \left\langle \frac{R}{6}(P_u^2 + \zeta Q_u^2) - \frac{3R}{2}[P_x^2 + \zeta(Q_x^2 - 4PQ_x + 2P^2)] \right\rangle. \tag{N-01}$$

The derivative coupling mixes cosine and sine. The real block $P \propto \cos$, $Q \propto \sin$ and its complement each have two pairs, giving four in the complete real opposite-wavevector block. On the chosen interval $\zeta \leq 4/9$ and $k^2 \geq 1$ make both kinetic and potential forms positive. The Hamiltonian is time dependent, so this is not conserved energy or an all-time stability result.

For $m \neq 0$, Hermite–theta scalar functions are combined with all tensor spins s. LRS coefficients and ladder rules preserve $L = n + \mathrm{sgn}(m)s$, giving finite blocks including lapse, shift, constraints and gauge transformations. The Heisenberg representation remains infinite dimensional, and no rotational Killing field on the fixed quotient is assumed.

Grade L	Complex metric	Constraint rank	Complex pairs
-2	1	0	1
-1	2	1	1
0	4	3	1
1	5	4	1
≥ 2	6	4	2

The table fixes $m > 0$ and one lattice copy r. Real fields including $-m$ double these counts, and all copies multiply them by m. At the edge, tensor and multiplier slots with $n < 0$ do not exist;

the remaining constraints have full row rank. Closure and the absence of nonzero-central KIDs give the generic $L \geq 2$ count. Explicit matrices are provided for $L = -2, -1, 0, 1, 2$, with an independent four-dimensional check at $L = 2$. Non-LRS coefficients retain $n \leftrightarrow n + 2$ couplings and do not close in this finite grading.

6.4 Sol: the Gowdy sector and nonzero dual orbits

In the fiber-independent sector, $P = 2x + p$ and $Q = e^{-2x}r$ make p, r periodic with $L_{\mathrm{Sol}} = \log[(3 + \sqrt{5})/2]$. For $k = 2\pi n/L_{\mathrm{Sol}} \neq 0$, one real quadrature has reduced action

$$L_k^{[2]} = \frac{t}{2}\left[\dot{p}^2 + \dot{r}^2 - k^2 p^2 - (k^2 + 4)r^2\right]. \tag{SO-01}$$

There are two propagating fields and four pairs in the complete real block. The coefficient $k^2 + 4$ belongs to the chosen areal time and variables, not a conserved physical frequency. A general nonzero fiber sector retains an entire dual-lattice monodromy orbit, whose coefficient is a function on the real line.

$$D_1 = \partial_\xi, \qquad D_2 = iK, \qquad D_3 = iM, \qquad K = k_0 e^{-\xi}, \quad M = l_0 e^{\xi}. \tag{SO-02}$$

All metric, lapse, shift, constraint and gauge operators preserve the same infinite orbit. Exponential multiplication couples infinitely many fixed Hermite coefficients. The differential right inverse in Appendix G has a denominator bounded away from zero on every nonzero orbit of the fixed lattice and reduces the system to two complex canonical field pairs. Including the conjugate orbit gives four real canonical field pairs. These count species in an infinite-dimensional orbit, not a finite total of two pairs.

6.5 Taub on the three-sphere: complete Wigner blocks

Expand invariant-frame components in Wigner functions, retaining all right indices at fixed j. Restoring left multiplicity and the real-field condition gives the following counts. The label j is not the round-sphere TT harmonic label.

j	Real constraint rank	Real pairs	KID deficiency
0	3	3	Axial $U(1)$
1/2	16	8	None
1	33	21	Left $SU(2)$
3/2	64	32	None

The $j = 0, 1$ values are linear gauge quotients at symmetric backgrounds, with quadratic KID conditions still to be imposed. Their mechanism differs from absent slots at the Nil edge. At $j = 3/2$ there are two fields with sixteen real Wigner components each, hence 32 pairs. Reconstruction with the gauge, time-dependent embedding and multipliers satisfies all four-dimensional linear vacuum components. The table covers only the four computed labels, not a uniform all-j classification.

6.6 Finite-time examples and an independent Sol comparison

For the initial data in Appendix H, the Sol Gowdy Bessel evaluation, first- and second-order Nil base evolutions, and reconstructed ADM evolutions for S^3, $j = 3/2$ and Nil $m = 1, r = 0, L = 2$

agree. These numerical comparisons verify nontrivial linear examples. Independently, (SL-01) gives existence of a local nonlinear lift for the admissible Sol initial data.

For Sol orbit $(1, 0)$ we evolved an independently derived four-dimensional Lorenz system from the same initial ADM data. On $1 \le t \le 2$, $|\xi - \xi_0| \le 4$, reduced-field L^2 differences were 3.26×10^{-8} for 1023 to 1535 nodes, 2.77×10^{-10} for the window comparison, and 3.94×10^{-9} for time refinement. Actual derivatives of the time interpolant give a maximum ten-component Einstein L^2 residual of 3.83×10^{-11} at twenty sampled times.

In contrast, substituting the reconstructed reduced variables into the reduced fourth-order system (SG-05) gives a maximum normalized strong residual (SH-05) of 1.54×10^{-3} in the same test. Evaluated at the saved times of the four trajectories, it lies between 9.6×10^{-4} and 4.5×10^{-3}; it does not decrease from 1023 to 1535 nodes and grows with the looser time tolerance, so convergence in this form is not established. The residual norms are different. Appendix H specifies finite-window Dirichlet conditions, interior characteristic margin and window controls. This example is not a whole-line error bound or a general stability result.

Figure 5 in Appendix H shows the reconstructed interior profiles.

7 Compact conditions and local integrability

7.1 Four flat-torus conditions and the existing theorem

Add a zero-mean TT field γ to the homogeneous volume velocity h and tracefree shear Σ of Paper06, and spatially integrate the original constraints to second order. Time translation and three spatial translations give

$$\begin{aligned} 6h^2 &= \operatorname{tr}(\Sigma^2) + \frac{1}{4}\left\langle \dot\gamma_{ij}^2 + q_0^{-2}(\partial_a \gamma_{ij})^2 \right\rangle, \\ 0 &= \langle \dot\gamma_{ij} \partial_a \gamma_{ij} \rangle \quad (a = 1, 2, 3). \end{aligned} \tag{W-03}$$

There are no further KIDs on the fixed flat torus. For smooth periodic linearized vacuum data on its boundaryless CMC Cauchy slice, Theorem 3.3 of Arms, Marsden and Moncrief makes (W-03) necessary and sufficient for tangency to a local exact vacuum family [18]. Appendix D specifies function spaces, KIDs and normalization of the theorem's cones.

Sufficiency comes from the existing theorem, not from a finite check of averages. Higher-order corrections may introduce new spatial harmonics. The statement does not address two-endpoint boundary data, global-in-time regularity or nonlinear stability. Lifting by the LC connection gives a local family in the same nondegenerate EC theory with constant complete NY coupling.

At $\gamma = 0$ one recovers the homogeneous cone (T-01). Positive energy excludes pure-wave directions with $h = \Sigma = 0$, while mixed directions with volume velocity can be nontrivial. A standing wave $A \cos x \cos t$ has $E = A^2$, zero translation charges and $h^2 = A^2/24$. A traveling wave $A \cos(x - t)$ has $E = 2A^2$, $P_x = -A^2$ and needs compensating momentum.

7.2 Explicit second-order standing-wave corrections

For $q_0 = 1$ and periods 2π, expand the exponents of the following metric as $\varepsilon(A_1, B_1, P_1) + \varepsilon^2(A_2, B_2, P_2)$.

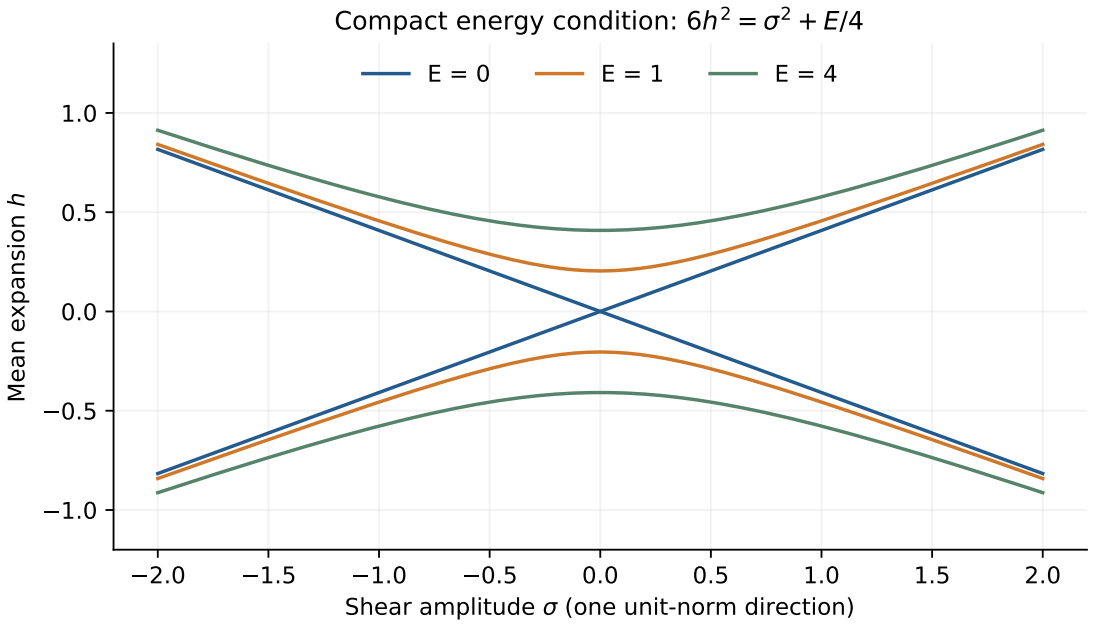


Figure 2: Energy section of (W-03). Together with the three translation conditions it is necessary and sufficient for local integrability under the stated flat-torus hypotheses.

$$
\begin{aligned}
ds^2 &= e^{2\mathcal{A}}(-dt^2 + dx^2) + e^{2\mathcal{B}+2\mathcal{P}}dy^2 + e^{2\mathcal{B}-2\mathcal{P}}dz^2, \\
A_1 &= B_1 = ht, \qquad P_1 = \frac{a}{2}\cos x \cos t, \qquad h^2 = \frac{a^2}{24}, \\
A_2 &= \frac{h^2t^2}{2} + \frac{a^2}{32}(1 - \cos 2t - \cos 2x), \\
B_2 &= -h^2t^2 - \frac{a^2}{64}\cos 2x \cos 2t, \\
P_2 &= \frac{ha}{2}(\sin t - t\cos t)\cos x.
\end{aligned}
\tag{W-04}
$$

For either sign of h, every first- and second-order Ricci coefficient vanishes. Zero, original and doubled wavevectors are retained; the truncated expression is not an exact finite-amplitude solution. For two nonparallel standing waves, $h^2 = (a^2 + b^2)/24$ and Appendix D constructs sources and corrections in all thirteen generated wavevectors. Polynomial and resonant time terms are retained, giving $O(\varepsilon^3)$ residuals on fixed finite intervals.

7.3 The central condition on Nil

The curved backgrounds (BF-01) are CMC with $K_{ij}K^{ij} > 0$. A compact integral identity forces the KID lapse to vanish. Spatial Killing fields descending to the fixed quotient give one central KID on Nil, none on Sol and four on Taub S^3. The central charge vanishes identically in the Nil $m = 0$ sector but is nontrivial for $m \neq 0$:

$$
\mathcal{Q}_Z = -2 \sum_{m>0,r,L} \mu \,\mathrm{Im}(\Pi^\dagger Q) = 0, \qquad \mu = \frac{m}{2\pi}.
\tag{K-01}
$$

Conjugate sectors add rather than cancel. The condition applies to the sum over blocks; individual charges need not vanish. For $L = 2$, $Q = (1, 0)$, $\Pi = (i, 0)$ satisfies linear constraints but has

charge 2μ. Nontrivial zero-charge linear examples also exist, without establishing nonlinear lifting sufficiency or second-order Einstein corrections here.

7.4 KIDs and periodic reconstruction on Sol

The fixed Sol quotient has no global KID, so neither the four flat-torus conditions nor the Nil central condition is added to a general orbit. Periodic reconstruction in the fiber-independent Gowdy representation involves the second-order mean correction $\langle \delta p_P^{(2)} \rangle$ of (KG-03). Fixing that correction to zero leaves a source condition on the first-order fields, not a common KID obstruction of the general fiber-dependent system. Nonlinear products generate zero modes and other orbits. The following proposition permits those corrections in the full vacuum theory; it does not assert nonlinear closure within a single orbit.

Proposition (local integrability on the fixed Sol background). Fix a slice $t = t_0 \in [1, 2]$ of (BF-01) on the fixed compact Sol quotient and an integer $s \geq 6$. Every real solution (h, p) of the full linearized vacuum constraints in $H^s \times H^{s-1}$, with p a momentum-density variation, is tangent to an exact vacuum constraint-data family:

$$\begin{aligned} &\Phi(g_\varepsilon, \pi_\varepsilon) = 0, \\ &(g_\varepsilon, \pi_\varepsilon) = (g_0, \pi_0) + \varepsilon(h, p) + O_{H^s \times H^{s-1}}(\varepsilon^2). \end{aligned} \tag{SL-01}$$

This applies the existing no-KID theorem [19]. Appendix G specifies the bundles, function spaces, surjectivity and quotient KIDs. Local Cauchy development and the LC connection give a local family in the same nondegenerate source-free EC theory with constant complete NY coupling.

The proposition includes nonzero-base Gowdy modes and real initial data reconstructed by (SG-04) from the common core of (SF-02), with the conjugate orbit restored. Infinite orbit sums require convergence in the reconstructed global Sobolev norm. Higher-order corrections may contain zero modes and other orbits; fixed mean curvature, the same reduced gauge and preservation of the Gowdy ansatz are not imposed on the family. Local time existence is distinct from a uniform guarantee on the numerical interval, well-posedness or numerical convergence of (SG-05), and nonlinear stability.

7.5 Four conditions on Taub three-spheres

$$\mathcal{Q}_X = \int_{S^3} p^{ij} \mathcal{L}_X h_{ij} \, d\mu = 0, \qquad X \in \mathfrak{su}(2) \oplus \mathfrak{u}(1). \tag{K-02}$$

The three left $SU(2)$ conditions act on left coefficient indices. The axial $U(1)$ condition includes tensor-frame rotation as well as the right derivative. The adjoint kernels at $j = 0, 1$ correspond to these KIDs. A finite-time $j = 3/2$ example satisfies local linear constraints and all four quadratic necessary conditions, without proving nonlinear lifting for arbitrary data, all-j estimates or horizon stability.

7.6 Domains of the conditions

Background	Global KIDs	Conclusion here
Fixed flat T^3	4	Necessary and sufficient for local integrability under the stated hypotheses
Nil LRS	1	Quadratic necessary central-charge condition
Fixed Sol	0	Local integrability in the stated Sobolev class, (SL-01); mode reconstruction conditions in Appendix G
S^3 Taub	4	Four quadratic necessary conditions

8 Conclusions

For nondegenerate EC theory with constant complete NY on fixed compact geometries, connection and metric descriptions agree through the action, constraints and endpoint data. The homogeneous regular-stratum counts remain 5/3/2/2; the static torus and symmetric Taub points require separate treatment. Three descriptions of nonzero flat modes agree in the two-polarization action and reduced two-form.

Under the stated Cauchy and regularity hypotheses, the existing theorem makes (W-03) necessary and sufficient for local flat-torus integrability. Single and nonparallel standing waves admit corrections satisfying the local equations through second order. Local existence of exact families, explicit finite-order formulas and long-time behavior remain distinct results.

On curved vacuum backgrounds we reduced central-zero and central-nonzero Nil blocks, the fiber-independent sector and infinite dual orbits on Sol, and low Wigner labels on S^3. Field species are distinguished from block totals, and Nil edge effects from Taub KID rank exceptions. Background-dependent quadratic necessary conditions and finite-time reconstructed examples were obtained.

On the fixed Sol background, absence of global KIDs and the elliptic splitting of the constraint map give local integrability for all real full linearized constraint data with integer $s \geq 6$. This includes the stated Gowdy data, real general-orbit data reconstructed from the common core, and the saved Hermite initial profile; infinite sums require (SL-07). Higher-order corrections may contain zero modes and other orbits. This existence result concerns the full vacuum theory on a short time interval.

The independent Lorenz example on Sol supports finite-time interior convergence, while its normalized strong residual in the reduced fourth-order system remains near 10^{-3} without convergence in that form. Whole-line guarantees, all-orbit/all-j/all-time stability, non-LRS Nil and general triaxial Bianchi IX, lifting sufficiency on Nil and Taub, closed realizations and well-posedness for arbitrary reduced Sol L^2 data, exact families confined to a fixed sector, and full classification of singular strata remain open.

Acknowledgments

The author acknowledges Claude Opus 4.8, Gemini 3.1 Pro, Grok 4.3, and GPT-5.5 Thinking via ChatGPT for assistance during preparation of the original manuscript in June 2026. For this revision, OpenAI Codex assisted with symbolic derivations, implementation, Japanese–English exposition and deterministic figure-generation code. The calculations and figures are reproducible

from the supplied mathematical definitions and data. The author takes responsibility for the content and remaining errors.

This work was developed within the informal collaborative project **DPPU** (**D**onut-like topology, **P**lanck-scale compactness, **P**recession dynamics, **U**niverse).

Availability and contact

The manuscript and supplement are available at https://github.com/Muacca/DPPUv7-paper07. The existing Zenodo record is doi:10.5281/zenodo.20678646; cite the edition and date of the artifact actually obtained. The supplement uses the MIT License. Contact: muacca@dmwp.jp.

A Fixed quotients and the convention dictionary

On T^3, x, y, z have period 2π and $\sigma = (dx, dy, dz)$. On Nil use $\sigma = (dx, dy, dz - xdy)$ with the following lattice generators.

$$(x, y, z) \sim (x + 2\pi, y, z + 2\pi y) \sim (x, y + 2\pi, z) \sim (x, y, z + 4\pi^2). \tag{QA-01}$$

Each generator preserves the coframe, and the fundamental volume is $(2\pi)^4$. The central Fourier factor is $e^{imz/(2\pi)}$, $m \in \mathbb{Z}$. The dual fields $X = \partial_x$, $Y = \partial_y + x\partial_z$, $Z = \partial_z$ give $-(X^2 + Y^2 + Z^2)$. The $m = 0$ sector is the base torus; $m \neq 0$ gives an oscillator with frequency $|m|/(2\pi)$ plus the central constant $m^2/(4\pi^2)$. Twisted periodicity gives $|m|$ independent states for each signed m.

For Sol take $\sigma = (dx, e^x dy, e^{-x} dz)$, $A = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and $\lambda = (3 + \sqrt{5})/2$. Choose a unit-area fiber lattice in eigenvector coordinates of A and identify (x, y, z) with $(x + \log\lambda, \lambda^{-1}y, \lambda z)$. Then $n_{23} = n_{32} = 1$ and the reference volume is $\log\lambda$. Fiber Fourier covectors are grouped into dual-lattice orbits under A.

For simply connected $S^3 = SU(2)$ use the unit round coframe satisfying $d\sigma^i = 2\sigma^j \wedge \sigma^k$ cyclically. The metric $q^2 \sum_i (\sigma^i)^2$ has sectional curvature q^{-2}, scalar curvature $6q^{-2}$ and volume $2\pi^2 q^3$. A structure matrix defined with a negative structure equation changes sign when converted to the present convention. Axis changes must act consistently, including orientation, on metric, connection and torsion.

$$C^i{}_{jk} = \frac{a}{r}\epsilon_{ijk} \quad \Longrightarrow \quad q = \frac{2r}{|a|}, \quad R = \frac{3a^2}{2r^2}, \quad \mathrm{Vol} = \frac{16\pi^2 r^3}{|a|^3}. \tag{QA-02}$$

Equation (QA-02) converts an orthonormal-frame structure magnitude a/r to physical radius. Thus $a = 1$ gives $q = 2r$, $a = 2$ gives $q = r$, and $a = 4$ gives $q = r/2$. Curvature must not be converted while volume or wave numbers are left unchanged. The present calculation uses geometric torsion T directly. The supplementary comparison guide gives implementation conversions.

$$\mathcal{N} = \frac{1}{4}[abcd]T^e{}_{ab}T_{ecd} - \frac{1}{2}[abcd]R_{abcd}, \qquad [0123] = +1. \tag{QA-03}$$

The alternating symbol here is not the epsilon tensor with indices raised by the Lorentz metric. Both terms of complete NY use the same connection and agree with the exterior derivative in (A-01).

A.1 Perturbation normalizations and symbols

In the inhomogeneous calculations a common positive action factor is suppressed and integration uses fixed Haar measure. Symmetric metric components are ordered $(11, 22, 33, 12, 13, 23)$, with conjugate momentum $\delta\pi^{ii}$ on the diagonal and $2\delta\pi^{ij}$ off the diagonal. Thus J is the standard canonical matrix. A dagger includes complex conjugation and the formal spatial adjoint.

$$z = (h, p), \qquad J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}, \qquad \Omega = -J, \qquad [X_i, X_j] = C^k{}_{ij} X_k. \tag{QA-04}$$

On Nil, m is the central Fourier integer, r a lattice copy, n the oscillator index and L the tensor grade. On S^3, j, m, n are Wigner labels, independent of the background Taub parameter fixed to zero. On Sol, x is the base coordinate and ξ the unfolded real coordinate. The Nil period is also denoted $P_0 = 2\pi$ to distinguish it from the Gowdy field P; the LRS background value of P in (NG-02) is $\bar{P}$ and $\zeta = e^{2\bar{P}}$. The Sol period is $L_{\mathrm{Sol}} = \log\lambda$, distinct from the Lagrangians L_0 and $L^{[2]}$ and from the Nil grade L.

B Full contortion, Euler equations and endpoints

$$\begin{aligned} K_{ab} &= K_{ab,c} e^c, \qquad K_{ab,c} = -K_{ba,c}, \qquad s_a = \eta_{aa}, \\ R_{\mathrm{EC}} &= R_{\mathrm{LC}} + \nabla_a J^a + R_K, \qquad J^a = 2 s_a \sum_b s_b K_{ab,b}, \\ R_K &= \sum_{a,b,e} s_a s_b s_e (K_{ae,a} K_{eb,b} - K_{ae,b} K_{eb,a}). \end{aligned} \tag{CB-01}$$

Use the 24 independent coordinates $a < b$, $c = 0, \ldots, 3$. The Hessian of R_K has determinant 2^{32} and rank 24; the 24×24 linear map from K to T is also invertible. The determinant depends on coordinate normalization, whereas nondegeneracy does not. The contraction proof of the Cartan equation in Section 2 gives uniqueness for general nondegenerate coframes; this component calculation is an algebraic control.

Equation (CB-01) is tensorial and applies also with shift. The spatial slice has no boundary, so the integrated divergence contributes only the normal current at time endpoints. Treating the LC GHY term, this current and constant NY in corresponding boundary representatives gives $S_0 = S_{\mathrm{ADM}} + \int \sqrt{-g}\, R_K/(2\kappa^2)$. Its K equation gives $K = 0$ pointwise.

$$\begin{aligned} \omega &= K + W(e, \partial e), \quad W = \omega_{\mathrm{LC}}, \\ E_K(S') &= E_\omega(S), \\ E_e(S') &= E_e(S) + E_\omega(S) \frac{\partial W}{\partial e} - \partial_\mu \left(E_\omega(S) \frac{\partial W}{\partial(\partial_\mu e)} \right). \end{aligned} \tag{CB-02}$$

Equation (CB-02) is the Euler chain rule with contractions suppressed. On the complete connection equation the extra terms vanish, and the coframe equation becomes the LC Einstein equation. This does not justify discarding selected Euler equations before imposing the connection shell.

The endpoint one-form acquires a term proportional to E_ω from the integration by parts in (CB-02); its pullback to the connection shell vanishes. Boundary representatives additionally differ by δB_{end}, whose exterior variation is zero. Hence the reduced two-forms agree. This is a statement about pullbacks to solution space and corresponding endpoint data, not a claim that the derivative-dependent change is a point canonical transformation.

$$
\begin{aligned}
L_0 &= c\left[-\frac{6q\dot{q}^2}{N} + 6\chi Nq - 6Nq\eta^2 + \frac{2}{3}Nq^3V^2\right], \\
B_{\mathrm{end}} &= -2cq^3V + 6\theta\mathcal{V}_\Sigma q^2\eta, \\
(p_\eta, p_V) &= (0,0), \qquad (\eta, V) = (0,0).
\end{aligned}
\tag{CB-03}
$$

In the common-scale family the last four conditions are four second-class constraints. In the endpoint-retaining representative the primaries become $P_\eta - \partial_\eta B = 0$, $P_V - \partial_V B = 0$, and $P_q = p_q + \partial_q B$. Transforming constraints and Hamiltonian by the same substitution preserves the Poisson algebra and Dirac two-form. The conclusion is not extended to variable θ, spin sources or degenerate coframes.

B.1 Direct checks in inhomogeneous blocks

$$
B_{\mathrm{EH}} = \tfrac{1}{2}\epsilon_{abcd}e^a \wedge e^b \wedge K^{cd}, \qquad B_{\mathrm{NY}} = e^a \wedge T_a, \qquad dB_{\mathrm{NY}} = T^a \wedge T_a - e^a \wedge e^b \wedge R_{ab}. \tag{CB-04}
$$

Coupling coefficients are suppressed in (CB-04). The EH current may be expressed in contortion coordinates while the NY current is expanded in raw connection perturbations; they must be put in the same connection variables before combination. At quadratic order we retain products of the two quadratures and, where required, the mean second-order Cartan contribution paired with the background.

For a nonzero flat mode the complete chains with 40 configurations including six Lorentz directions, 34 after Lorentz gauge fixing, and 10 metric configurations were compared directly. Nil base modes and Sol Gowdy modes include direct checks of all 24 Cartan components, the auxiliary quadratic form and endpoint currents. At $j = 3/2$ on S^3, all right-index Cartan components and the quadratic endpoint identity were checked. For central Nil and general Sol orbits the Cartan solution respects the tensor/orbit closure; these checks are distinct from computing additional complete Dirac chains or full quadratic endpoint matrices.

C Constraint ranks and route controls

A constant symmetric n can be diagonalized by $SO(3)$. The A_k in (H-03) represent the adjoint structure algebra; the Jacobi identity yields the momentum Poisson algebra. Both T and P are invariant under this action, giving $\{J, F\} = 0$. In the six coordinates and their conjugates these statements are directly checkable polynomial identities.

$$
\begin{aligned}
g &= \mathrm{diag}(1,2,3), \quad c = 1, \quad M = \pi g, \\
M_{T^3} &= \mathrm{diag}(1,1,0), \quad M_{\mathrm{Nil}} = \sqrt{3}I, \\
M_{\mathrm{Sol}} &= \frac{5\sqrt{3}}{3}I, \quad M_{S^3} = \mathrm{diag}(4\sqrt{2},0,0).
\end{aligned}
\tag{CC-01}
$$

All representatives satisfy $F = J = 0$. For T^3 the minor is $\partial F/\partial p_{33} = -6$. For the others take the required initial rows of (F, J_1, J_2, J_3) and columns of $(p_{11}, p_{23}, p_{13}, p_{12})$. Their determinants are $9\sqrt{3}$, $50\sqrt{3}$, and $-64\sqrt{2}$ for Nil, Sol and S^3, respectively; the Sol value uses the diagonalized $n = \mathrm{diag}(0,1,-1)$. Thus the constraint Jacobian ranks are 1/3/4/4.

The spans of the A_k have dimensions 0/2/3/3, bounding the ranks of J. Near the nonzero minors, these maximal ranks and the independence of dF persist, defining the regular strata of (H-04). Counts are not extended uniformly to points where minors vanish, such as symmetry-enhanced configurations.

C.1 Closure for quadratic actions

$$M_{Ab} = \{\Phi_A, \phi_b\}, \quad u^A M_{Ab} = 0 \quad \Longrightarrow \quad u^A\{\Phi_A, H_c\} \simeq 0. \tag{CC-02}$$

Here ϕ denotes the original primaries and Φ all independent constraints available at the current stage. Add only new independent linear constraints from the left nullspace of M, iterating until none remain. For constant-coefficient quadratic actions this linear-algebra procedure is exact. It is not used to replace nonlinear constraints by linear-coefficient ranks. If the final Poisson rank is s and there are m independent constraints, their first- and second-class counts are $m - s$ and s.

Control	Connection: FC/SC/pairs	Eliminated: FC/SC/pairs
Complete torus quadratic	1 / 4 / 1	1 / 0 / 1
Complete sphere quadratic	0 / 8 / 0	0 / 4 / 0

For the eliminated sphere block, $H = -p_u^2/24 - 6\ell u$ and $p_\ell = 0$ give $u = 0$, then $p_u = 0$, then $\ell = 0$. On the torus the lapse primary is first class and the quadratic action retains a scalar pair. Section 5 treats the condition from the original nonlinear constraint.

C.2 Scalar controls from the complete action

Set $c = 1$, $n = 2kI$ and write the independent scalar connection as $\omega^0{}_i = be^i$, $\omega^j{}_k = ae^i$ cyclically. Here $k = 0$ is the torus and $k = 1$ the round-sphere reference.

$$L_{\mathrm{P}} = 6Nq^3b^2 - 12q^2\dot{q}\,b + 12kNq^2a - 6Nq^3a^2, \quad a = k/q, \quad b = \dot{q}/(Nq). \tag{D-02}$$

$$\begin{aligned} L_{\mathrm{met}} &= 6k^2Nq - 6q\dot{q}^2/N, \\ q &= 1 + \varepsilon u, \quad N = 1 + \varepsilon\ell, \quad a = k + \varepsilon A, \quad b = \varepsilon B, \\ L^{[1]} &= 6k^2(\ell + u), \\ L_{\mathrm{P}}^{[2]} &= -6A^2 - 12kAu + 6B^2 - 12B\dot{u} + 6k^2\ell u - 6k^2u^2, \\ L_{\mathrm{met}}^{[2]} &= -6\dot{u}^2 + 6k^2\ell u. \end{aligned} \tag{D-03}$$

Substitution of the quadratic connection equations $A = -ku$, $B = \dot{u}$ gives the same metric action. The sphere has a nonzero first variation; the zero-pair agreement is an off-shell algebraic control. The torus quadratic scalar action retains one pair, while the original lapse equation requires $\dot{u}^2 = 0$, the pure-scale section of the cone in Section 5.

C.3 Counts at symmetric Taub points and nonzero modes

At the Taub point (BF-01), the homogeneous momentum and total ranks are two and three, leaving three pairs in the linear quotient. This is a different stratum from the regular representative (CC-01), with ranks three and four. Appendix G gives the $j = 0, 1$ KID exceptions and nonzero minors valid on the whole interval. The 40/34/10-configuration flat comparison in Section 4 counts one real Fourier quadrature, separately from the regular homogeneous count (H-04).

D Fourier action and integrated constraints

$$\Gamma^{i(1)}{}_{jk} = \tfrac{1}{2}(\partial_j h_{ik} + \partial_k h_{ij} - \partial_i h_{jk}), \quad R^{[1]} = \partial_i\partial_j h_{ij} - \partial^2 h_{ii}. \tag{WD-01}$$

The second coefficient of the integrated $\sqrt{g}R$ Gamma–Gamma form gives the spatial term of (W-01). The first lapse perturbation multiplies $R^{[1]}$. With the wavevector along z, $R^{[1]} = k^2(h_{11}+h_{22})$; the shift enters through v, retaining all constraints. The momentum equations are proportional to $p_{13} = p_{23} = p_{33} = 0$ and the linear Hamiltonian condition to $h_{11} + h_{22} = 0$. These four secondaries and four primaries mutually Poisson commute.

For $k \neq 0$, quotienting gauge directions and taking $h_{11} = -h_{22} = A$, $h_{12} = B$ gives (W-02). For any nonzero lattice wavevector the same algebra holds in an orthonormal polarization basis. The basis rotation need not be a lattice automorphism: the original wavevector and polarization tensors are restored.

$$\begin{aligned} R + K^2 - K_{ij}K^{ij} = 0, \qquad D_j(K^{ij} - g^{ij}K) = 0, \\ K^{[1]} = hI + \Sigma + \tfrac{1}{2}\dot{\gamma}, \qquad \left\langle R^{[2]} \right\rangle = -\tfrac{1}{4q_0^2}\left\langle (\partial_a \gamma_{ij})^2 \right\rangle . \end{aligned} \tag{WD-02}$$

Zero-mean TT fields have vanishing averaged cross terms with constant h, Σ. The linear curvature of the second-order metric correction integrates to zero as a divergence, and because the background K vanishes, the second-order correction to K does not enter the quadratic constraint linearly. The averaged Hamiltonian constraint is therefore $6h^2 - \mathrm{tr}\Sigma^2 - \langle\dot{\gamma}^2\rangle/4 - \langle(\partial\gamma)^2\rangle/(4q_0^2) = 0$.

$$\mathcal{J}[\xi] = \int_\Sigma \pi^{ij}\mathcal{L}_\xi g_{ij}, \quad \xi = \partial_a : \qquad \mathcal{J}^{[2]}[\partial_a] \;\propto\; \langle \dot{\gamma}_{ij}\partial_a\gamma_{ij}\rangle . \tag{WD-03}$$

Because background $\pi = 0$ and g is constant, linear terms in the second-order fields vanish, leaving (WD-03). Cross terms with constant mean metric or shear integrate to zero by periodicity. This gives the three momentum conditions in (W-03). The homogeneous linear equations keep h, Σ constant, while the wave equation and periodic integration conserve wave energy and momentum.

The integrated conditions are the compact linearization-stability conditions [17, 18]. Positive TT energy excludes pure-wave directions but permits mixed directions with volume velocity. We next state their sufficiency on the fixed flat torus and give explicit second-order corrections.

D.1 Application of the known theorem on the fixed flat torus

$$\partial_i\partial_j\nu - \delta_{ij}\Delta\nu = 0, \qquad \partial_i X_j + \partial_j X_i = 0, \qquad \det(\mathcal{S}_k^\dagger \mathcal{S}_k) = 32(k^2)^5 \quad (k \neq 0). \tag{WD-04}$$

In units $q_0 = 1$, (WD-04) describes the adjoint kernel. Taking the trace and using periodicity makes ν constant; the periodic Killing equation makes X constant. These exhaust the four KIDs. Coordinate-linear rotations and boosts do not descend to the fixed lattice. The last identity checks the combined four-column symbol at nonzero wavevector.

$$\begin{aligned} \delta g = C + 2t(hI + \Sigma) + \gamma, \qquad & \delta\pi = \Sigma - 2hI + \tfrac{1}{2}\dot{\gamma}, \\ \beta = \mathrm{tr}(\delta\pi) = -6h, \qquad & \omega^{\mathrm{TT}} = \Sigma + \tfrac{1}{2}\dot{\gamma}, \\ \langle(\omega^{\mathrm{TT}})^2\rangle - \beta^2/6 + \tfrac{1}{4}\langle(\partial\gamma)^2\rangle = \mathrm{tr}\Sigma^2 - 6h^2 + \tfrac{1}{4}\langle\dot{\gamma}^2 + (\partial\gamma)^2\rangle. & \end{aligned} \tag{WD-05}$$

C and Σ are constant, Σ is tracefree and γ is mean-zero TT. Equation (WD-05) identifies the Hamiltonian cone, Eq. (3.5) of Arms–Marsden–Moncrief, with the energy condition in (W-03); their momentum cones, Eq. (3.1), agree with (WD-03). Their Section 3 treats exactly a flat g_0 with $\pi_0 = 0$, and their introduction includes the flat spacetime $T^3 \times \mathbb{R}$ with four Killing fields. We apply their Theorem 3.3 (Section 3, p. 96) to smooth periodic data on a compact boundaryless CMC slice. Curves may be interpreted, conservatively, in $H^s \times H^{s-1}$ for integer $s \geq 5$, without claiming minimal regularity. The four conditions then characterize tangents to

exact constraint-data curves. Local Cauchy development and the LC lift give EC vacuum families [18].

This concerns a local initial-value family on a common time interval, not a two-time boundary-value problem, all-time regularity or an exact family closed in a finite Fourier space. Higher-order corrections may contain new harmonics.

D.2 Explicit second-order corrections and two-wave closure

For the standing wave (W-04), expanding each exponent as $\epsilon A_1 + \epsilon^2 A_2$, and similarly for B, P, makes the first and second coefficients of all ten Ricci components vanish, for either sign of h. This is a local substitution check through second order, not an exact finite-amplitude solution.

$$\begin{aligned} f &= a\cos x\cos t, \quad g = b\cos y\cos t, \quad h^2 = (a^2+b^2)/24, \\ H_{\mu\nu} &= \mathrm{diag}(-6ht,\, 2ht+g,\, 2ht+f,\, 2ht-f-g), \\ \mathcal{K} &= \{(0,0),(\pm1,0),(0,\pm1),(\pm2,0),(0,\pm2),(\pm1,\pm1)\}. \end{aligned} \tag{WD-06}$$

$$\begin{aligned} \ddot{\bar{Q}}_{\mu\nu,n} + |n|^2\bar{Q}_{\mu\nu,n} &= -2G^{[2]}_{\mu\nu,n}[H], \\ -\dot{\bar{Q}}_{0\nu} + \partial_i\bar{Q}_{i\nu} &= 0, \\ \psi &= -\tfrac{7}{16}(a^2\cos 2x + b^2\cos 2y) - ab\cos x\cos y, \\ \Delta\psi &= -2G^{[2]}_{00}[H](0), \\ \bar{Q}_{ij}(0) &= \delta_{ij}\psi, \qquad \bar{Q}_{00}(0) = \bar{Q}_{0i}(0) = 0, \\ \dot{\bar{Q}}_{ij}(0) = \dot{\bar{Q}}_{00}(0) &= 0, \qquad \dot{\bar{Q}}_{0i}(0) = \partial_i\psi. \end{aligned} \tag{WD-07}$$

Write $g = \eta + \epsilon H + \epsilon^2 Q$, with bars denoting trace reversal by η. Source conservation and the initial conditions in (WD-07) preserve the second-order harmonic condition. Keeping all thirteen wavevectors, integration of the forced oscillators constructs Q satisfying the local Einstein equations and reality through second order. The public `two_waves.json` supplies the source and correction coefficients. Removing the zero wavevector does not give the same construction.

E Geometric comparisons

This appendix collects geometric data used to interpret the constraint counts and wave analysis. Closed forms and fixed quotients follow Paper04, the frame comparison follows Paper06, finite Nil metrics are distinguished by the integral harmonic forms of Paper03EC, and the Sol isometric path follows Paper04 [4, 5, 7]. The formulas are assembled using the same lattices, radii and action normalization.

E.1 Closed one-forms and frame representations

$$\alpha = v_i\sigma^i, \quad v_i \text{ constant}: \qquad d\alpha = 0 \iff nv = 0. \tag{G-02}$$

In the specified invariant coframes the constant closed-one-form dimension is $3 - \operatorname{rank} n$. These forms are also coclosed on the compact unimodular quotient. Invariant forms compute degree-one cohomology for standard Nil and completely solvable Sol; together with T^3, S^3 this gives $b_1 = 3/2/1/0$ [21, 22]. This identification does not extend to constant-component forms in an arbitrary frame.

The first Betti numbers can also be checked from the lattice abelianizations. On Nil, the commutator of the x, y generators is the central generator, leaving a two-dimensional real

abelianization. On Sol one keeps the base generator and the fiber coinvariants under $A - I$; since A has no eigenvalue one, the real fiber contribution vanishes and $b_1 = 1$. Together with $b_1(T^3) = 3$ and simple connectivity of S^3, the closed forms in the specified coframes represent all of H^1 for these four quotients.

The flat-torus example of Paper06, Section 7 and Appendix A, takes the following form in the present convention.

$$\begin{aligned} \sigma^1 &= \cos(\nu z)\,dx + \sin(\nu z)\,dy, \\ \sigma^2 &= -\sin(\nu z)\,dx + \cos(\nu z)\,dy, \qquad \sigma^3 = dz, \quad \nu \in \mathbb{Z} \setminus \{0\}, \\ \sum_i (\sigma^i)^2 &= dx^2 + dy^2 + dz^2, \qquad n = \mathrm{diag}(-\nu, -\nu, 0). \end{aligned} \tag{G-03}$$

On the same flat torus with 2π periods, the coordinate frame has rank zero and (G-03) has rank two. Their constant closed-form dimensions are three and one, while $b_1 = 3$ is unchanged. The forms dx, dy acquire spatially dependent coefficients in the rotated frame. This separates a local frame gauge transformation of the same fields from different constant-component truncations. Rank invariance under constant $SO(3)$ conjugacy remains valid.

E.2 Finite metric paths on fixed quotients

$$g = q_0^2\,\mathrm{diag}(e^{2\beta_1}, e^{2\beta_2}, e^{2\beta_3}), \qquad \sum_i \beta_i = 0. \tag{G-04}$$

$$\begin{aligned} e &= E\sigma, \quad E = q_0\,\mathrm{diag}(e^{\beta_i}), \quad m = EnE^T/\det E, \\ \omega^0{}_i &= 0, \qquad \omega^i{}_j = \epsilon_{kij}\left[\tfrac{1}{2}(\mathrm{tr}m)\delta_{k\ell} - m_{k\ell}\right] e^\ell, \\ S(s) - S(0) &= cq_0^3 \Delta\tau\,[R(s) - R(0)]. \end{aligned} \tag{GE-03}$$

In (GE-03), s is a finite parameter labeling static metrics, not a time-dependent field. The lattice and volume $q_0^3\mathcal{V}_\Sigma$ are fixed, and the LC connection is specified along every path. Torsion and the NY current vanish, as does extrinsic curvature at the time endpoints, so the action difference over a proper-time interval $\Delta\tau$ is proportional to the scalar-curvature difference. Solving the connection equation is distinct from solving the metric vacuum equation.

Path	$(\beta_1, \beta_2, \beta_3)$	$q_0^2 R$
T^3 shape	$(s/2, -s/2, 0)$	0
Nil flat	$(s/2, -s/2, 0)$	$-1/2$
Nil vertical	$(-s/2, -s/2, s)$	$-e^{4s}/2$
Sol base	$(s/3, -s/6, -s/6)$	$-2e^{-2s/3}$
Sol isometric	$(0, s, -s)$	-2

The Sol paths use the coframe with $n_{23} = n_{32} = 1$. The Nil-vertical and Sol-base derivatives are nonzero for every finite s, so their path curvature is not a mass at an extremum. Nil flat has constant scalar curvature but a nonzero full shape gradient and is not a vacuum modulus. The Sol-isometric path is generated by base translation descending to the fixed mapping torus.

$$\delta g = 2q^2 B, \quad B^T = B, \quad \mathrm{tr}B = 0: \qquad \delta R = -\frac{2}{q^2}\mathrm{tr}(XB). \tag{GE-01}$$

Equation (GE-01) follows directly by differentiating the scalar-curvature formula. Since $X \neq 0$ for Nil and Sol, stationarity fails in the full shape space. The DeWitt norm is positive on tracefree B, but does not turn a nonstationary path Hessian into a physical mass.

$$\left(\left\langle \frac{dx}{2\pi}, \frac{dx}{2\pi} \right\rangle, \left\langle \frac{dy}{2\pi}, \frac{dy}{2\pi} \right\rangle\right) = \frac{\mathcal{V}_\Sigma q_0}{4\pi^2}(e^{-s}, e^{s}). \tag{G-05}$$

For Nil flat, $dx/(2\pi), dy/(2\pi)$ form an integral harmonic basis. The shortest nonzero integral squared norm is the common coefficient in (G-05) times $e^{-|s|}$, distinguishing compact metrics with different $|s|$. A discrete identification can relate s and $-s$. This distinguishes finite metrics on the fixed quotient without claiming vacuum stationarity or a complete moduli classification.

$$f_s(x, y, z) = (x + s, y, z), \qquad f_s^*\sigma = (\sigma^1, e^s\sigma^2, e^{-s}\sigma^3). \tag{GE-02}$$

The map f_s commutes with the fiber translations and monodromy identification of Sol, so it descends to the fixed compact quotient. The Sol-isometric path is $f_s^* g_0$ and does not define a new modulus. This contrasts with the integral harmonic-lattice invariant along Nil flat.

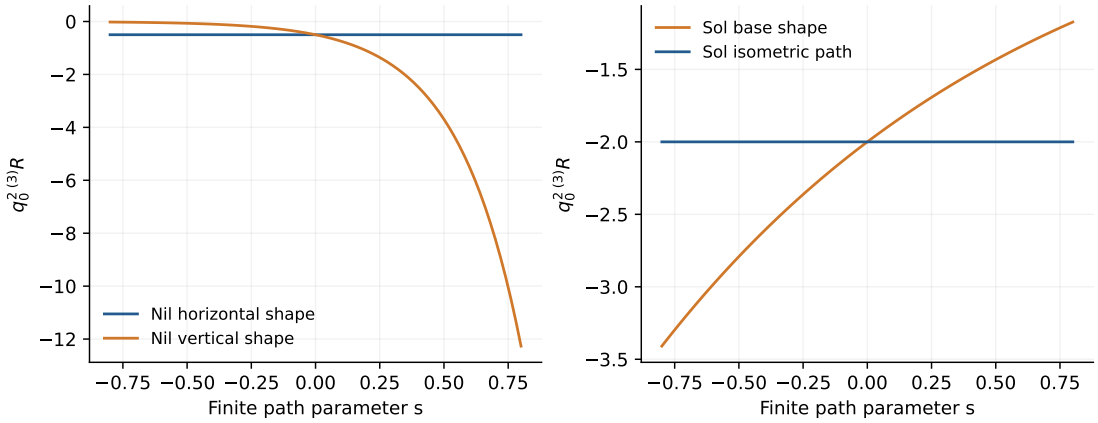


Figure 3: Scalar curvatures of finite paths on fixed quotients, providing geometric comparisons for the metric constraints in the main text.

E.3 Reference geometric operators

The following are geometric operators on the specified static Riemannian metrics. Their operators and background conditions are recorded separately from the physical torus modes derived from the action in Sections 6–7.

$$\lambda_{\mathrm{rough,TT}}(S^3) = \frac{\ell(\ell+2) - 2}{q^2}, \qquad \ell \geq 2, \qquad d_\ell = 2(\ell-1)(\ell+3). \tag{S-01}$$

Equation (S-01) is the rough-Laplacian spectrum on rank-two TT tensors of the round S^3 [23]. It is neither the scalar Laplacian nor the Lichnerowicz operator. The static sphere is off shell in the present model, so these eigenvalues are not assigned physical vacuum frequencies.

$$\begin{aligned} &\lambda_{\mathrm{tor}}(\mathrm{Nil}) = (m^2 + n^2)/q^2, \quad (m, n) \in \mathbb{Z}^2, \\ &\lambda_{k,j}(\mathrm{Nil}) = q^{-2}\left[\frac{k^2}{4\pi^2} + \frac{|k|}{2\pi}(2j+1)\right], \quad k \neq 0, \quad j \geq 0. \end{aligned} \tag{S-02}$$

The Nil formula is the scalar Laplacian on the lattice of Appendix A. Each signed k has multiplicity $|k|$, with coincident eigenvalues from distinct labels combined separately. The eigenvalue includes both the central k^2 term and the horizontal oscillator contribution [24].

$$L_{p_y,p_z} = q^{-2}\left[-\frac{d^2}{dx^2} + p_y^2 e^{-2x} + p_z^2 e^{2x}\right], \qquad L_{p_y,p_z} \geq \frac{2|p_y p_z|}{q^2}. \tag{S-03}$$

For Sol, (S-03) applies to each nonzero dual-lattice orbit, with domain determined by the monodromy orbit identification [25]. The bound follows from the arithmetic–geometric mean inequality for the potential and nonnegativity of the kinetic term. The zero orbit is a separate base-circle sector. These scalar operators provide reference comparisons; the coupled gravitational operators on the selected vacuum backgrounds are treated in Section 6 and Appendices F–H.

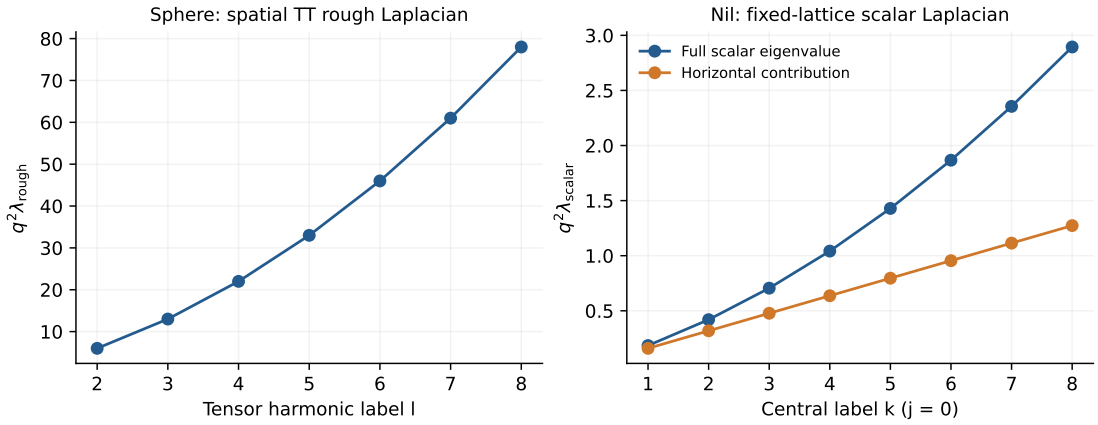


Figure 4: The spherical TT rough Laplacian and Nil scalar Laplacian, in the radius and lattice conventions used here.

F Vacuum backgrounds, fixed quotients and harmonic bases

F.1 Background construction and descent to the quotients

The Nil background in (BF-01) is Tanimoto's homogeneous vacuum solution with $b = 1$, Kasner exponents $(2/3, 2/3, -1/3)$ and literature time u^3. Sol is the twisted Gowdy solution $P = 2x, Q = 0, R = t$. The biaxial Taub background uses $\ell = 1, m = 0$ and the conversion $\sigma = -\omega/2$. Direct substitution gives zero for the full Ricci tensor, both ADM constraints and the metric/momentum evolution residuals at the stated lapse. Time reparametrization does not change the lattice [26, 27, 28].

$$\begin{aligned}
\pi^{ij}_{\mathrm{Nil}} &= \mathrm{diag}(-4/(9A), -4/(9A), -u^2),\\
\pi^{ij}_{\mathrm{Sol}} &= \mathrm{diag}(-1/F, -t - 1/(4t), -t - 1/(4t)),\\
\pi^{ij} &= \sqrt{\det G}\,(K^{ij} - G^{ij}K), \qquad K_{ij} = \dot{G}_{ij}/(2N).
\end{aligned} \tag{BF-02}$$

The three Nil deck generators preserve the coframe in (QA-01). The Sol gluing preserves $(e^x dy, e^{-x} dz)$, and the S^3 coframe is global. Consequently the time-only G, N descend to the fixed quotients. Positivity on the closed intervals follows directly from (BF-01).

F.2 Central-nonzero Nil basis and tensor grade

$$\begin{aligned} \mu &= m/P_0, \quad m \in \mathbb{Z} \setminus \{0\}, \quad 0 \le r < |m|, \quad P_0 = 2\pi, \\ \Phi_{m,r,n} &= \sqrt{P_0}\, e^{i\mu z} \sum_{k\in\mathbb{Z}} e^{i(r+km)y} f_n\!\left(x + \frac{r+km}{\mu}\right), \\ f_n(v) &= \frac{(|\mu|/\pi)^{1/4}}{\sqrt{2^n n!}} H_n(\sqrt{|\mu|}v) e^{-|\mu|v^2/2}. \end{aligned} \tag{NF-01}$$

Use normalized measure $dx\,dy\,dz/P_0^4$. The lattice shift in x reindexes the sum, and the y, z periods give integer phases, so (NF-01) descends to the quotient. Distinct r, m are orthogonal. Unfolding the sum of x intervals to the real line reduces normalization to that of f_n.

$$\begin{aligned} (X - iY)\Phi_n &= \sqrt{2\mu n}\,\Phi_{n-1}, \qquad (X + iY)\Phi_n = -\sqrt{2\mu(n+1)}\,\Phi_{n+1}, \quad Z\Phi_n = i\mu\Phi_n \quad (m > 0), \\ \theta^{\pm} &= (\sigma^1 \pm i\sigma^2)/\sqrt{2}, \qquad \theta^0 = \sigma^3, \qquad L = n + \mathrm{sign}(m)s. \end{aligned} \tag{NF-02}$$

The symmetric tensor basis is $\theta^+\theta^+$, $(\theta^+\theta^0 + \theta^0\theta^+)/\sqrt{2}$, $(\theta^+\theta^- + \theta^-\theta^+)/\sqrt{2}$, $\theta^0\theta^0$ and the two negative-spin partners, with spins $2, 1, 0, 0, -1, -2$. For $m > 0$ only components with $n = L - s \ge 0$ exist. The LRS background has spin zero; ladder moves and tensor-spin moves preserve L. Applying the same rule to metric, momentum, lapse, shift and Cartan tensors closes a finite block. Complex conjugation gives $m < 0$, with $r \mapsto -r \bmod |m|$ and $s \mapsto -s$.

The spin is a local rotation grading, not an additional global rotation KID of the quotient. Non-LRS coefficients couple n to $n \pm 2$ and destroy this finite-grade closure; this does not exclude other non-LRS analyses. At $m = 0$ the usual base Fourier basis applies.

F.3 Fixed Sol lattice and unfolding

$$\begin{aligned} \varphi &= (1 + \sqrt{5})/2, \quad \lambda = \varphi^2, \quad L_{\mathrm{Sol}} = \log\lambda, \\ S &= \frac{1}{\sqrt{1+\varphi^2}} \begin{pmatrix} 1 & -\varphi \\ \varphi & 1 \end{pmatrix}, \quad D = \mathrm{diag}(\lambda^{-1}, \lambda), \quad SA = DS, \\ q &= 2\pi S(n_1, n_2)^T = (k_0, l_0)^T, \qquad q_j = (D^T)^j q, \\ k_0 l_0 &= \frac{4\pi^2}{\sqrt{5}}(n_1^2 - n_1 n_2 - n_2^2) \ne 0 \quad (n \ne 0). \end{aligned} \tag{SF-01}$$

The fiber lattice is $S\mathbb{Z}^2$ with area one. The integer quadratic form in (SF-01) cannot vanish for a nonzero integer pair, so every nonzero dual orbit has both fiber components. Continuous covectors on a fiber axis are not modes of this fixed quotient.

$$\begin{aligned} U_q f &= \sqrt{L_{\mathrm{Sol}}} \sum_{j\in\mathbb{Z}} f(x + jL_{\mathrm{Sol}}) e^{i(\lambda^{-j}k_0 y + \lambda^j l_0 z)}, \\ D_1 &= \partial_\xi, \quad D_2 = iK, \quad D_3 = iM, \qquad K = k_0 e^{-\xi}, \quad M = l_0 e^{\xi}, \\ \mathcal{D} &= \{f : e^{a|\xi|}\partial_\xi^b f \in L^2(\mathbb{R}) \text{for all integers } a, b \ge 0\}. \end{aligned} \tag{SF-02}$$

For normalized measure $dx\,dy\,dz/L_{\mathrm{Sol}}$, U is an isometry from $L^2(\mathbb{R})$ onto the orbit space, by deck reindexing and fiber orthogonality. All coefficients, derivatives and formal adjoints are treated on $\mathcal{D}$, with zero boundary terms. Smooth compactly supported functions and Hermite functions belong to this core. The passage from this core to smooth quotient constraint data and local integrability is proved in (SL-06); it does not provide a closed realization or a well-posedness theorem for the fourth-order reduced system.

$$[D_1, D_2] = -D_2, \quad [D_1, D_3] = D_3, \quad e^{\xi}\psi_0 = e^{1/4}\sum_{n=0}^{\infty} \frac{\psi_n}{2^{n/2}\sqrt{n!}}. \tag{SF-03}$$

Every coefficient in (SF-03) is nonzero, so no finite Hermite truncation is exactly closed. The infinite orbit is closed under all tensor components, constraints, gauge and evolution. Its conjugate orbit completes a real field. Fiber-zero modes instead use periodic base Fourier functions and form a distinct sector.

F.4 Wigner blocks and reality on the three-sphere

$$\begin{aligned} Y^j_{mn} &= \sqrt{2j+1}\, D^j_{mn}, \quad d = 2j+1, \quad \overline{Y^j_{mn}} = (-1)^{m-n} Y^j_{-m,-n}, \\ D_a &= 2iJ_a, \quad [D_a, D_b] = -2\epsilon_{abc} D_c, \quad \sum_a D_a^2 = -4j(j+1)I. \end{aligned} \tag{TF-01}$$

Normalize Haar measure by the volume. Expand each of the six metric coefficients in the background coframe over all right indices. The left index supplies multiplicity; a complete real block at fixed j contains d^2 real scalar components. Tensor coefficients obey the corresponding conjugation condition in (TF-01). We classify only $j = 0, 1/2, 1, 3/2$ explicitly and use $j = 3/2$ for complete reconstruction.

G Constraint operators, reduction and reconstruction

G.1 Common construction

$$\begin{aligned} \mathcal{H}[N] &= \int N \left[\frac{\pi^{ij}\pi_{ij} - \pi^2/2}{\sqrt{g}} - \sqrt{g}\, R \right], \\ \mathcal{H}[X] &= \int \pi^{ij} \mathcal{L}_X g_{ij}, \qquad Cz = \delta(\mathcal{H}_\perp, \mathcal{H}_i), \qquad H = \delta^2 \mathcal{H}[N], \\ S^{[2]} &= \int dt\, [p^\dagger \dot{h} - \tfrac{1}{2} z^\dagger H z - \alpha^\dagger C z]. \end{aligned} \tag{RG-01}$$

Complex sectors are paired with their conjugates to give a real action. The Hamiltonian row of C includes the background lapse because its multiplier is $\delta N/N$. Vary (RG-01) in the fixed frame, retaining its structure constants in Lie and covariant derivatives. For example, $\delta\Gamma = G^{-1}(\nabla h + \nabla h - \nabla h)/2$ and $\delta\mathrm{Ric} = \nabla\delta\Gamma - \nabla\delta\Gamma$ generate the spatial Hessian. These formulas define the public matrices reproducibly.

$$\begin{aligned} CJC^\dagger = 0, \quad \dot{C} + CJH = BC, \quad CE = GE = 0, \quad E^\dagger \Omega E = \Omega_r, \\ H_r = E^\dagger H E - E^\dagger \Omega \dot{E}, \quad \dot{z} = JHz + JC^\dagger \alpha, \\ \alpha = -(GJC^\dagger)^{-1}(\dot{G}E + GJHE)y, \qquad z = Ey. \end{aligned} \tag{RG-02}$$

Select independent constraint rows at constant rank and impose G. The time derivative of E in (RG-02) is essential on an evolving background, even when its contribution vanishes in a particular gauge. An independent four-dimensional Einstein variation checks the reconstructed spatial equations and time components proportional to constraints. The public coefficients expand and check this derivation.

G.2 Nonzero base modes on Nil

$$
\begin{aligned}
Gz = 0: \quad & h_{12} = h_{13} = 0, \quad h_{22}/A + h_{33}/B_0 = 0, \quad 2A\delta\pi^{11} + 2\pi_0^{11} h_{11} = 0, \\
& P = h_{33}/B_0, \quad Q = h_{23}/B_0, \quad B_0 = 16u^2/(9A), \\
& \Pi_P = -A\delta\pi^{22} + B_0\delta\pi^{33} + (A\pi_0^{22} + B_0\pi_0^{33})P, \\
& \Pi_Q = B_0 p_{23} + 2B_0\pi_0^{22} Q.
\end{aligned}
\tag{NG-01}
$$

$$
\begin{aligned}
P_{uu} + u^{-1}P_u + 9(k^2 + 2\zeta)P - 18ik\zeta Q = 0, \\
Q_{uu} + (u^{-1} + 2\bar{P}_u)Q_u + 9k^2 Q + 18ikP = 0, \\
\bar{P} = \log\frac{4u}{3(1+u^4)}, \quad \zeta = e^{2\bar{P}}, \quad R = 4u/3, \qquad k \neq 0.
\end{aligned}
\tag{NG-02}
$$

These expressions align the base wavevector with one axis; the corresponding polarization rotation is restored for a general integer base direction. The chosen constraint minor is $-128ik^3u^5(1+u^4)/9$ and the gauge minor is $256ik^3u/81$, nonzero throughout the interval. Substitution of (NG-01) in (RG-02) gives (N-01) and then (NG-02). Off-diagonal spatial terms mix the cosine and sine quadratures. The interval bound $k^2 - 2\zeta \geq 1/9$ does not define a time-independent conserved energy.

G.3 Central-nonzero Nil blocks

Expand (RG-01) in the basis (NF-02), including only multipliers present at each grade. Solve selected independent momentum columns from the constraints and fix their conjugate metric columns by gauge. Every existing constraint row has full rank. With $b = \sqrt{\mu} > 0$, explicit minors at $L = -1, 0, 1, 2$ are as follows; $L = -2$ has no constraint row.

$$
\begin{aligned}
M_{-1} &= 2b(1+u^4), \qquad M_0 = -256b^3u^5/9, \\
M_1 &= -\frac{512b^3u^3[9b^2u^2(1+u^4)^2 + 12u^4 + 4]}{81(1+u^4)}, \\
M_2 &= -2048b^3u^5(1+u^4)/9.
\end{aligned}
\tag{NG-03}
$$

For general $L \geq 2$ there are six metric components and four multipliers. Finite closure identifies an adjoint kernel with a KID in the same central-nonzero sector, whereas the only quotient KID is the central-zero field Z. Thus all four constraints are independent, leaving two complex pairs. This structural argument is distinct from explicit matrices at five grades. The lower-end reduction comes from absent $n < 0$ components, not a KID rank defect.

$$
H_{r,-2} = -2\begin{pmatrix} \dfrac{81\mu^2A^2 - 144\mu u^2 + 64}{72uA^2} & \dfrac{2(u^4-1)}{uA} \\ \dfrac{2(u^4-1)}{uA} & \dfrac{9A^2}{2u} \end{pmatrix}.
\tag{NG-04}
$$

Equation (NG-04) gives the one-pair lowest endpoint explicitly. Its sign in a time-dependent gauge does not by itself classify ghosts or instability. At $L = 2$ the reduction, gauge multipliers and all ten Einstein components were independently compared; this four-dimensional check is not claimed uniformly for every grade.

G.4 Fiber-zero Gowdy sector on Sol

$$
\begin{aligned}
ds^2 &= -N^2dt^2 + e^{2a}(dx+\beta dt)^2 + R[e^P(dy+Qdz)^2 + e^{-P}dz^2],\\
R &= t+\rho, \quad P = 2x+p, \quad Q = e^{-2x}r, \quad e^{2a_0} = F, \quad k = 2\pi n/L_{\mathrm{Sol}},\\
C_H &= \delta p_R + (t-1/(4t))\delta p_a + (2-1/(2t^2)-2k^2)\rho + 2tikp,\\
C_x &= -2ik\delta a + ik(-2t-1/(2t))\rho + 2\delta p_P - ik\delta p_a.
\end{aligned}
\tag{SG-01}
$$

For $n \neq 0$, the gauge $\rho = \delta p_a = 0$ has determinant $-2ik$. Eliminating $\delta a = \delta p_P/(ik)$ and $\delta p_R = -2tikp$ gives (SO-01). The two equations are $\ddot{p}+t^{-1}\dot{p}+k^2p = 0$ and $\ddot{r}+t^{-1}\dot{r}+(k^2+4)r = 0$. The auxiliary metric variable obeys $\ell = 4t\dot{p}/(ik)$ and $\dot{\ell} = 4iktp$.

G.5 Differential right inverse for a general nonzero Sol orbit

The following are differential identities on the entire orbit, not inverses of a finite truncation. Transform h, p by $U = \mathrm{diag}(1,1,1,i,i,1)$ and the constraint rows by $\mathrm{diag}(1,1,-i,-i)$ to obtain real coefficients. Primes denote ξ derivatives. Write the momentum part of $C = (C_q, C_p)$ as $C_p p = r$.

$$
\begin{aligned}
r_0 &= A_0p_{11} + p_{22} + p_{33}, \quad A_0 = F(4t^2-1)/(2t),\\
r_1 &= -2Fp'_{11} + 2t(p_{22}-p_{33}) + F(Kp_{12}+Mp_{13}),\\
r_2 &= -t(p'_{12} + p_{12} + 2Kp_{22} + Mp_{23}),\\
r_3 &= -t(p'_{13} - p_{13} + 2Mp_{33} + Kp_{23}).
\end{aligned}
\tag{SG-02}
$$

$$
\begin{aligned}
f &= p_{11}, \quad b = Kp_{12} - Mp_{13}, \quad V = K^2+M^2, \quad d_0 = 2 - FV/(2t),\\
a &= \frac{-b' - (K^2-M^2)r_0 + A_0(K^2-M^2)f - Vr_1/(2t) - FVf'/t - Kr_2/t + Mr_3/t}{d_0},\\
p_{22} &= (r_0 - A_0f)/2 + r_1/(4t) + Ff'/(2t) - Fa/(4t),\\
p_{33} &= (r_0 - A_0f)/2 - r_1/(4t) - Ff'/(2t) + Fa/(4t),\\
p_{12} &= (a+b)/(2K), \quad p_{13} = (a-b)/(2M),\\
p_{23} &= (-r_2/t - p'_{12} - p_{12} - 2Kp_{22})/M.
\end{aligned}
\tag{SG-03}
$$

Using (SF-01), $F/t \geq e$ and $V \geq 2|k_0l_0|$ gives $d_0 \leq 2 - 4\pi^2e/\sqrt{5} < 0$ for every nonzero lattice orbit on $1 \leq t \leq 2$. Define E_p by setting $r = 0$, and R_p by setting $f = b = 0$. With $TQ = (Q_1, 0, 0, KQ_2, -MQ_2, 0)$, one obtains the complete splitting below.

$$
\begin{aligned}
C_pE_p = 0, \quad C_pR_p = I, \quad T^\dagger E_p = I, \quad T^\dagger R_p = 0,&\\
E_pT^\dagger + R_pC_p = I, \qquad Gz = R_p^\dagger h = 0,&\\
E = \begin{pmatrix} T & 0 \\ -R_pC_qT & E_p \end{pmatrix}, \quad Q = E_p^\dagger h, \quad \Pi = (p_{11}, Kp_{12} - Mp_{13}),&\\
GJC^\dagger = I, \quad GJG^\dagger = 0, \quad E^\dagger\Omega E = \Omega_r.&
\end{aligned}
\tag{SG-04}
$$

The gauge is equivalent to $h_{22} = h_{33} = h_{23} = 0$ and $h_{12}/K + h_{13}/M = 0$. Here $E^\dagger\Omega\dot{E}$ vanishes exactly, but $\dot{E}$ remains in reconstruction. If R is the reconstruction discrepancy, constraint and gauge preservation give $CR = GR = 0$, while the reduced equation gives $E^\dagger\Omega R = 0$. The complete splitting (SG-04) places R in the range of E; nondegeneracy of Ω_r gives $R = 0$. This proof does not require brute-force cancellation of every expanded reconstruction entry.

$$H_r = \sum_{a=0}^{4} A_a(t,\xi)\partial_\xi^a, \qquad H_r^\dagger = H_r, \qquad \operatorname{rank} A_4 = 1, \qquad (J_r A_4)^2 = 0. \tag{SG-05}$$

The public supplement expands all coefficients in (SG-05) and checks the highest-order rank and nilpotence. These facts alone do not imply hyperbolicity or stability.

G.6 Low-label ranks and reconstruction on the three-sphere

Per fixed left index, the constraint ranks at $j = 0, 1/2, 1, 3/2$ are $3, 8, 11, 16$. Rational polynomial root counts, endpoints and denominators certify nonzero minors throughout $\tau \in [1/4, 1/2]$. Restoring left multiplicity and reality gives the complete-block counts in the text. At $j = 0$ the two-form on the ℓ, m parameter variations is $\left(\begin{smallmatrix} 0 & -64 \\ 64 & 0 \end{smallmatrix}\right)$; time displacement is pure gauge.

$$\begin{aligned} T_q &= \operatorname{diag}(1,1,1,i,1,i) \otimes \operatorname{diag}(1,\sqrt{3},\sqrt{3},1), \\ Gz = 0: \quad & h_{11} = h_{22} = h_{33} = h_{13} = 0 \quad \text{for all right indices}, \\ \det M &= \frac{2^{52}\tau^4(\tau-1)^4(\tau+1)^4(\tau^4-6\tau^2+33)^2(3\tau^4+14\tau^2+3)^2}{81(1+\tau^2)^4}. \end{aligned} \tag{TG-01}$$

Equation (TG-01) gives the constant basis transformation and the selected minor at $j = 3/2$; momenta transform by the inverse adjoint. With $w = (1, \sqrt{3}, \sqrt{3}, 1)$, the eight free pairs are $h_{12}/(iw), h_{23}/(iw)$ and $-iwp_{12}, -iwp_{23}$. Constructing E and the multipliers by (RG-02), independent four-dimensional variation gives $\delta G_{ij} = 0$, $\delta G_{00} = -NC_H/(2\sqrt{g})$ and $\delta G_{0i} = -NC_i/(2\sqrt{g})$. Including all left indices gives 32 real pairs.

G.7 KIDs and quadratic necessary conditions

$$\int_\Sigma \left(|D\nu|^2 + K_{ij}K^{ij}\nu^2 \right) dV = 0 \quad \Longrightarrow \quad \nu = 0. \tag{KG-01}$$

The selected backgrounds are CMC with $K_{ij}K^{ij} > 0$, so (KG-01) applies to the KID lapse. The remaining spatial Killing fields must descend to the fixed quotient: one central field Z on Nil, none on Sol, and four from $SU(2) \times U(1)$ on Taub. The central-nonzero Nil charge is (K-01); the conjugate $-m$ contribution adds with the same sign. The condition is imposed on the sum over blocks and allows inter-block cancellation. At $m = 0$, Z acts trivially.

$$L_a = -D_a^T, \qquad B_3 = I_6 \otimes D_3 + S_3 \otimes I_d, \qquad (S_3 h)_{ij} = -C^l{}_{3i} h_{lj} - C^l{}_{3j} h_{il}. \tag{KG-02}$$

The three left charges on S^3 are $\operatorname{Re}\sum \bar{p} L_a h$, and the axial charge is $\operatorname{Re}\sum \bar{p} B_3 h$. Both terms in (KG-02) are required: the right scalar derivative alone omits the coframe spin. These are quadratic necessary conditions from (K-02).

$$\begin{aligned} \langle 2\delta p_P + \varepsilon\left[\delta p_P p_x + \delta p_r (r_x - 2r)\right]\rangle &= 0, \\ \langle \delta p_P^{(1)} \rangle = 0, \qquad 2\langle \delta p_P^{(2)} \rangle + \left\langle \delta p_P^{(1)} p_x + \delta p_r (r_x - 2r) \right\rangle &= 0. \end{aligned} \tag{KG-03}$$

Periodic metric reconstruction in Sol Gowdy requires the x-mean $\langle\cdot\rangle$ of $\Pi_P P_x + \Pi_Q Q_x$ to vanish, where Π_P, Π_Q are the momenta of P, Q in (SG-01). With p, r and their momenta $\delta p_P, \delta p_r$ of order ε, this mean equals ε times the first line of (KG-03). The second line expands $\delta p_P = \delta p_P^{(1)} + \varepsilon \delta p_P^{(2)} + O(\varepsilon^2)$ and fixes the homogeneous second-order correction by first-order fields. It is not a KID charge of a general orbit. For Nil and Taub, vanishing of the displayed charges alone is not used to conclude nonlinear lifting sufficiency. On Sol, sufficiency follows from the function spaces and theorem application below.

G.8 Local integrability on Sol

G.8.1 Background and all global KIDs

The quotient (SF-01) is the mapping torus of $A \in SL(2,\mathbb{Z})$ acting on the fiber torus, hence a connected, oriented, closed compact three-manifold. The coframe $(dx, e^x dy, e^{-x} dz)$ is global. Background (BF-01) is smooth for all $t > 0$, with positive lapse and spatial metric. In a product neighborhood of any $t_0 \in [1,2]$, compactness and locally bounded causal speeds make each slice Cauchy. We use the vacuum verification of Appendix F and

$$\begin{aligned} K^i{}_j &= \frac{1}{\sqrt{F}}\mathrm{diag}\left(t - \frac{1}{4t}, \frac{1}{2t}, \frac{1}{2t}\right), \\ \tau := \operatorname{tr} K &= \frac{t + 3/(4t)}{\sqrt{F}}, \qquad |K|^2 = \frac{(t - 1/(4t))^2 + 1/(2t^2)}{F} > 0. \end{aligned} \tag{SL-02}$$

For a KID (f, X), taking the trace evolution, including the inverse-metric variation with our positive sign for K, gives $0 = \Delta f - |K|^2 f + X(\tau)$ on the vacuum constraints, where $\Delta = \mathrm{div}\nabla$. The CMC background has $X(\tau) = 0$, so integration by parts gives (KG-01) and $f = 0$. The remaining equations are $\mathcal{L}_X g = \mathcal{L}_X K = 0$.

The spatial Ricci endomorphism has eigenvalues $(-2/F, 0, 0)$. A Killing flow preserves the unit vector in the simple negative eigendirection, so $[X, \partial_x] = 0$. Solving the remaining Killing equations gives all local fields on the universal cover as $a(\partial_x - y\partial_y + z\partial_z) + b\partial_y + c\partial_z$. Descent through the two fiber lattice translations forces $a = 0$; the base deck transformation then requires $(b, c) = (\lambda^{-1} b, \lambda c)$, hence $b = c = 0$. Thus there are no global KIDs. Discrete isometries need not be absent.

G.8.2 Function spaces and surjectivity

Fix an integer $s \geq 6$, and let Dens denote real densities. Sobolev norms and formal-adjoint pairings are fixed using the smooth background g_0:

$$\begin{aligned} \mathcal{X}_s &= H^s(S^2T^*\Sigma) \times H^{s-1}(S^2T\Sigma \otimes \mathrm{Dens}), \\ \mathcal{Y}_s &= H^{s-2}(\mathrm{Dens}) \times H^{s-2}(T^*\Sigma \otimes \mathrm{Dens}), \\ \Phi(g, \pi) &= (\mathcal{H}_\perp, \mathcal{H}_i), \qquad P = D\Phi_{(g_0, \pi_0)} = (P_g, P_\pi). \end{aligned} \tag{SL-03}$$

Here Φ consists of all Hamiltonian and momentum constraints in (RG-01). On the open set of positive metrics it is a smooth map $\mathcal{X}_s \to \mathcal{Y}_s$. Densities are identified using the fixed background volume density. The map $\pi^{ij} = \sqrt{g}(K^{ij} - g^{ij}\tau)$ and its inverse are smooth in this class; p denotes $\delta\pi$. We use the function-space setting of [19, p. 155], without a minimal-regularity claim.

The Hamiltonian row has orders two in g and zero in π; the momentum rows have at most order one in each. With a positive second-order operator B on the momentum-density variation bundle $S^2T\Sigma \otimes \mathrm{Dens}$, define the auxiliary spatial elliptic operator

$$\begin{aligned} B &= I + \nabla^*\nabla, \qquad \mathscr{L} = P_g P_g^* + P_\pi B P_\pi^*, \\ \sigma_4(\mathscr{L})(\xi) &= 2|\xi|^4 \mathrm{diag}\left(1, I_3 + |\xi|^{-2}\xi \otimes \xi\right), \\ \langle w, \mathscr{L} w\rangle &= \|P_g^* w\|^2 + \|P_\pi^* w\|^2 + \|\nabla P_\pi^* w\|^2. \end{aligned} \tag{SL-04}$$

The norms are background L^2 norms and $\xi \neq 0$ is a spatial covector. In a background orthonormal frame the Hamiltonian principal-symbol adjoint is, up to sign, $f(\xi \otimes \xi - |\xi|^2 I)$; the momentum symbol in π is $p \mapsto -2ip(\xi, \cdot)$. The off-diagonal blocks have order at most three, giving the positive

fourth-order symbol displayed in (SL-04). For smooth w, $\ker \mathscr{L} = \ker P^*$. A distributional solution of $P^*w = 0$ satisfies $\mathscr{L}w = 0$ and is smooth by elliptic regularity, so the KID argument also eliminates weak kernels.

Self-adjoint elliptic Fredholm theory on the closed manifold makes $\mathscr{L}$ invertible and gives the bounded right inverse and splitting

$$\begin{aligned} \mathscr{R} &= \begin{pmatrix} P_g^* \\ B P_\pi^* \end{pmatrix} \mathscr{L}^{-1} : \mathcal{Y}_s \longrightarrow \mathcal{X}_s, \qquad P\mathscr{R} = I, \\ \mathcal{X}_s &= \ker P \oplus \operatorname{range} \mathscr{R}, \qquad \operatorname{proj}_{\ker P} = I - \mathscr{R}P. \end{aligned} \tag{SL-05}$$

The inverse $\mathscr{L}^{-1}$ maps H^{s-2} to H^{s+2}; the two components of $\mathscr{R}$ have orders at most two and three, respectively, giving precisely $\mathcal{X}_s$. Thus P is surjective with closed range and split kernel. This implements the elliptic step in Theorem 2.1 of [19]. The auxiliary spatial operator $\mathscr{L}$ is distinct from the Hamiltonian of the fourth-order evolution system (SG-05).

G.8.3 Exact families, local development and the LC lift

Set $x_0 = (g_0, \pi_0)$ and consider $\Phi(x_0 + u + \mathscr{R}v) = 0$ for $u \in \ker P$, $v \in \mathcal{Y}_s$. Its derivative in v is the identity. The implicit function theorem gives $v = \psi(u)$ with $\psi(0) = D\psi(0) = 0$. For every $z \in \ker P$, the smooth initial-data curve $x_\varepsilon = x_0 + \varepsilon z + \mathscr{R}\psi(\varepsilon z)$ satisfies (SL-01). CMC was used to exclude background KIDs; fixed mean curvature is not imposed on all nearby data. Smoothness refers to the constraint zero set, not to the global quotient by all diffeomorphisms including discrete stabilizers.

Local Cauchy theory in harmonic or wave-map gauge gives vacuum developments on a common short time interval for sufficiently small ε [20]. Constraint and gauge propagation recover the full vacuum equations, with uniqueness up to diffeomorphism. The developments lie in $C^0H^s \cap C^1H^{s-1}$; tangency may be interpreted one derivative lower, in $C^0H^{s-1} \cap C^1H^{s-2}$. No smoothness of the evolution map in the top Sobolev norm is required. Choosing a coframe from the positive square root of the nearby spatial metric and adding its LC connection gives the local EC family by (A-02). The connection contains one derivative of the coframe. This Cauchy problem does not impose fixed data at two time endpoints.

G.8.4 Domains of the individual modes

For nonzero-base Gowdy modes, $\delta a = \delta p_P/(ik)$, $\delta p_R = -2tikp$ in (SG-01) and full metric–momentum reconstruction give smooth quotient constraint data. Restore the opposite-wavevector conjugate for a real field. Besides finite sums, one may take zero-mean periodic $p, r \in H^s$ and $\delta p_P, \delta p_r \in H^{s-1}$: inverse differentiation gains one order for δa, while δp_R uses one derivative of p. This division is not applied at $k = 0$. Allowing the second-order mean correction in (KG-03) is consistent with (SL-01).

For a general orbit, the isometry (SF-02) and the global frame imply

$$\begin{aligned} X_i U_q f &= U_q D_i f, \qquad U_q\mathcal{D} \subset \bigcap_{m\geq 0} H^m(\Sigma) \subset C^\infty(\Sigma), \\ E(t_0)\mathcal{D}^4 &\subset \mathcal{D}^{12}, \qquad CE = 0. \end{aligned} \tag{SL-06}$$

Every finite word $D_{i_1} \cdots D_{i_m}$ is a finite sum of exponentially weighted derivatives, hence maps $\mathcal{D}$ into L^2. Equivalence of frame-derivative and Sobolev norms on the closed manifold proves the first line; compact-support cutoff approximation in the weighted norms justifies weak derivative interchange. The denominator of (SG-03) stays away from zero, and each reconstruction coefficient

and each of its derivatives grows at most exponentially, so E preserves $\mathcal{D}$. Together with $CE = 0$, restoration of the original tensor phases and addition of the conjugate orbit, this puts the real data in the domain of (SL-01).

The finite Hermite sum (SH-03) is polynomial times Gaussian and belongs to $\mathcal{D}$, so its continuous whole-line initial profile is included. This does not certify a finite-window numerical trajectory as an exact solution. For multiple orbits, require, after imposing the real-field condition,

$$\sum_{[q]} \left(\|h_{[q]}\|^2_{H^s(\Sigma)} + \|p_{[q]}\|^2_{H^{s-1}(\Sigma)} \right) < \infty. \tag{SL-07}$$

Distinct orbits are orthogonal and frame derivatives preserve them, so the sum converges in $\mathcal{X}_s$ and satisfies all constraints by continuity of P. Smoothness of each orbit separately, or an unweighted square sum of reduced coefficients, does not imply (SL-07). Arbitrary reduced L^2 data, closed realizations, well-posedness and numerical convergence of (SG-05), all-time stability and nonlinear closure within a single sector remain outside this result.

H Finite-time examples and numerical comparisons on Sol

H.1 Initial data and checks for finite blocks

For Sol Gowdy take $n = 1$ and initialize at $t = 1$ from $p = J_0(kt)$ and $r = J_0(\sqrt{k^2 + 4}\, t)/2$ and their derivatives. For Nil base modes use wavevector $(1, 0)$, $P = \sqrt{2}p \cos x$, $Q = \sqrt{2}q \sin x$ and $(p, p_u, q, q_u) = (0.2, 0.07, -0.1, 0.03)$ at $u = 1$. Initial Hamiltonian momenta are obtained from (N-01).

For central Nil take $m = 1, r = 0, L = 2$, with $Q = 0$, $\Pi = (0.03, -0.02)$ at $u = 1$. For S^3, excite left $m = +3/2$ at $j = 3/2$ with $Q = 0$, first momentum 0.03 and sixth momentum -0.02 among the eight entries, at $\tau = 1/4$, and add its real-field conjugate at left $m = -3/2$. Other entries initially vanish; all right indices and reconstructed metric components are retained.

DOP853 runs compare relative tolerances 10^{-10} and 10^{-12}, with absolute tolerance smaller by a factor of 100, at 201 times on each interval. The table reports maximum component differences for the finer run, not rigorous error bounds.

Example	Independent comparison	Maximum difference	Constraint
Nil base	Second order / Hamilton	1.13×10^{-12}	3.89×10^{-16}
Nil central	Reduced / full ADM	7.30×10^{-13}	6.94×10^{-17}
Sol Gowdy	Numerical / Bessel	2.61×10^{-12}	Analytic reconstruction
S^3 Taub	Reduced / full ADM	1.45×10^{-13}	2.22×10^{-15}

The central Nil charge and all four S^3 charges vanish for these initial data and remain zero to numerical precision. These comparisons support finite-time linear examples. These numerical comparisons do not construct an exact nonlinear family. Local integrability of the admissible Sol initial data follows independently from the theorem application (SL-01).

H.2 Independent four-dimensional Lorenz system on Sol

$$\begin{gathered} \mathcal{F}_\nu = \nabla^\mu h_{\mu\nu} - \tfrac{1}{2}\nabla_\nu h^\mu{}_\mu, \\ \delta R_{\mu\nu} - \nabla_{(\mu}\mathcal{F}_{\nu)} = 0, \qquad H_{tt} = H_{\xi\xi} + B_x H_\xi + B_t H_t + B_0 H. \end{gathered} \tag{SH-01}$$

Direct Ricci variation in the noncoordinate frame of $\mathrm{diag}(-F, F, t, t)$ gives (SH-01). We use the standard Lorenz hyperbolic reduction, without WKB or geometric-optics approximations [29]. The ten components are ordered $(00, 01, 02, 03, 11, 22, 33, 12, 13, 23)$ with phases $(1, 1, i, i, 1, 1, 1, i, i, 1)$ to make coefficients real.

$$\begin{aligned} H_{00}(1) = H_{0i}(1) = 0, \quad (h,p)(1) = E(1)y(1), \\ \mathcal{F}(1) = \dot{\mathcal{F}}(1) = 0, \qquad y = -J_r E^\dagger J z, \quad (-J_r E^\dagger J)E = I, \\ (-J_r E^\dagger J) J C^\dagger = 0. \end{aligned} \tag{SH-02}$$

Match the spatial $H, \dot{H}$ to the same constrained ADM initial data, and determine $\dot{H}_{00}, \dot{H}_{0i}$ from the Lorenz condition. The independent wave equation and ADM constraints also give $\dot{\mathcal{F}}(1) = 0$ exactly. The projection (SH-02) removes gauge directions before comparison with the reduced variables y of Appendix G; raw metric components in different gauges are not equated. All B matrices and initial maps are supplied in the public supplement, implementing (RG-01) and direct Ricci variation.

$$\begin{aligned} &(n_1, n_2) = (1, 0), \quad \xi_0 = \tfrac{1}{2}\log(k_0/l_0), \\ &y_\alpha(1, \xi) = 8^{1/4} \sum_{n=0}^{31} c_{\alpha n} \psi_n(\sqrt{8}(\xi - \xi_0)), \qquad \alpha = 1, \dots, 4. \end{aligned} \tag{SH-03}$$

The 128 coefficients are fixed in `data/initial/sol_profile.json`, with a checked hash. They define the common initial profile. The profile is fixed input and is not interpreted as a conserved physical mode or frequency. Here ψ_n is the normalized Hermite function; neither the orbit nor the basis is changed with time.

H.3 Fixed discretization and three convergence comparisons

$$\xi_j = \xi_0 - W + \frac{2Wj}{N+1}, \quad j = 1, \dots, N, \qquad H(t, \xi) = \sum_{n=1}^{N} a_n(t) \sin \frac{n\pi(\xi - \xi_0 + W)}{2W}. \tag{SH-04}$$

Use the time-fixed sine basis (SH-04), homogeneous Dirichlet conditions at its endpoints and DST spatial derivatives. These boundary conditions do not guarantee constraint preservation. On $|\xi - \xi_0| \le 4$, $1 \le t \le 2$, continuum characteristic speed one leaves a margin of 0.25 for $W = 5.25$. The nonlocal spectral truncation is separately checked by changing the window; continuum causality alone does not imply zero numerical boundary error.

Comparison	Parameters	Maximum reduced L^2 difference
Space	$N = 1023 \to 1535$, $W = 5.25$	3.2645×10^{-8}
Window	$W = 5.25 \to 5.6875$, $N = 1535 \to 1663$	2.7682×10^{-10}
Time	rtol $10^{-12} \to 3 \times 10^{-14}$, $N = 1535$	3.9359×10^{-9}

Spatial and window comparisons use relative tolerance 3×10^{-14} and absolute tolerance 3×10^{-16}. The window comparison keeps the grid spacing fixed. Fields at 101 saved times are interpolated to the same 2048-point interior Gauss–Legendre quadrature. The relative tolerance approaches double precision; the table reports numerical comparisons for this implementation.

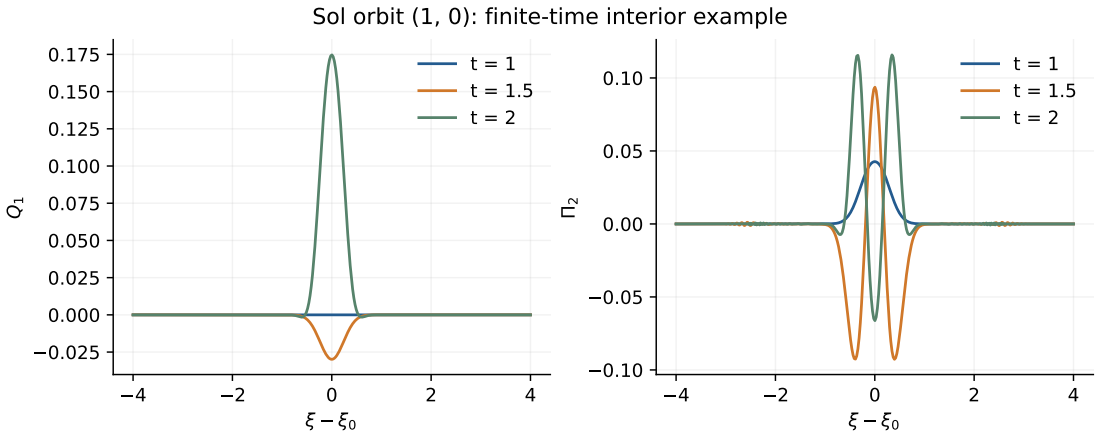


Figure 5: Reconstructed Q_1 and Π_2 on the interior from the same Sol initial profile. These are reduced variables in a finite-time example, not a gauge-independent metric observation or physical spectrum.

H.4 Continuous-time residuals and unresolved scope

$$\|v\|_I = \left(\int_{\xi_0-4}^{\xi_0+4} \sum_a |v_a|^2 d\xi \right)^{1/2}, \qquad \rho_{\mathrm{red}} = \frac{\|\dot{y} - J_r H_r y\|_I}{\max(1, \|J_r H_r y\|_I)}. \tag{SH-05}$$

Analytically differentiate the actual accepted seventh-degree DOP853 dense polynomial at twenty times $t = 1.025 + 0.05j$, $j = 0, \ldots, 19$, to obtain $H, \dot{H}, \ddot{H}$. These derivatives are substituted directly into the independent Ricci operator and the reduced fourth-order system (SG-05); $\ddot{H}$ is not replaced by the wave equation in (SH-01). Polynomial values are also compared directly with SciPy interpolation.

Diagnostic	Maximum on the interior
Lorenz wave equation	5.8631×10^{-10}
All ten Einstein components	3.8255×10^{-11}
Lorenz constraint	9.4853×10^{-15}
ADM constraint	2.4617×10^{-12}
Reduced fourth-order ρ_{red}	1.5388×10^{-3}

The first four rows are unnormalized L^2 norms of fixed-frame components; the last is the normalized residual (SH-05) of the reduced fourth-order system. Their magnitudes are not directly comparable error criteria. At the twenty times $\|J_r H_r y\|_I$ lies between 0.13 and 2.7, so the denominator in (SH-05) switches between absolute and relative normalization. The same residual is also evaluated on the four trajectories of the convergence comparisons at the 21 saved times $t = 1, 1.05, \ldots, 2$, with $\ddot{H}$ given by the wave equation in (SH-01).

Trajectory	Maximum ρ_{red} at the saved times
$N = 1023$, $W = 5.25$, rtol 3×10^{-14}	9.7624×10^{-4}
$N = 1535$, $W = 5.25$, rtol 3×10^{-14}	1.3726×10^{-3}
$N = 1663$, $W = 5.6875$, rtol 3×10^{-14}	9.6257×10^{-4}
$N = 1535$, $W = 5.25$, rtol 10^{-12}	4.5461×10^{-3}

From $N = 1023$ to $N = 1535$ the residual does not decrease, and with the looser time tolerance it grows by a factor of about three. Convergence in the reduced fourth-order form is therefore

not established. The independent Lorenz comparison supports a finite-time interior example. Whole-line error bounds, all-orbit or all-time stability and physical spectra remain unestablished. The continuous initial profile (SH-03) is covered by (SL-01), which does not certify exactness or convergence of the finite-window numerical trajectory.

I Revision note

This version uses geometric torsion, constant complete NY, the physical spherical radius and fixed compact quotients. The epsilon-form action coefficient and spherical normalization of the first edition are corrected, and the vector divergence and endpoint data are treated within one action. Sections 2 and Appendices A–B specify the conventions; the changelog records the changes between editions.

The identification of constant closed-one-form dimensions with physical scalar kernels, the common physical TT spectrum asserted for all four geometries, and the interpretation of the former DBR/RBD difference as a physical ordering preference are withdrawn. Primary preservation requires the multiplier term of (D-01), and quadratic comparisons require matching perturbative order. Geometric closed forms, finite paths and Laplacians are retained in the domains specified in Appendix E.

$$\begin{aligned} L_{\mathrm{old}} &= -8s\dot{b} + 12b^2 - 12a^2 - 4as - 12\ell a - 4\ell s, \\ L_{\mathrm{old,red}} &= -\tfrac{4}{3}\dot{s}^2 + 3\ell^2 - 2\ell s + \tfrac{1}{3}s^2, \\ \ell &= s/3, \qquad H_{PP} = -3/8. \end{aligned} \tag{D-04}$$

Equation (D-04) is the strictly quadratic trace block of the lowest sphere harmonic in the first edition. Connection elimination and complete primary preservation both give one pair and the same H_{PP}. Appendix C gives the scalar comparison derived from the complete action. The supplementary comparison guide supplies the legacy-variable embedding and its relation to the 26-variable block.

This version derives the regular constraint counts for all symmetric homogeneous metrics and the action reduction of nonzero torus modes, connecting the homogeneous cone of Paper06 to compact conditions with TT energy and translation charges; under the hypotheses of Appendix D these conditions are sufficient for local integrability by the Arms–Marsden–Moncrief theorem. It also reduces linear perturbations on selected Nil, Sol and Taub vacuum backgrounds, with quadratic KID conditions and finite-time examples (Appendices F–H). For the fixed Sol background, we additionally apply the no-KID theorem of Fischer, Marsden and Moncrief to establish local integrability of all full linearized constraint data in the specified Sobolev class, (SL-01)–(SL-07). This does not change the reservations about fourth-order reduced numerical convergence or long-time stability. Geometric results used from Papers 03EC, 04 and 06 are attributed and collected in Appendix E. Assumptions and derivations are given in the paper and appendices, with supplement reproduction instructions in the README.

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